

AKADEMIKA

FOL-M-GP

**FIBER OPTIC TRAINER KIT FOR
GLASS AND PLASTIC FIBER**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. FIBER OPTIC COMMUNICATION is one of the leading subjects of this century & expanding its horizon in coming century too.

Whether fiber optic system are for transmitting voice, video & audio or training regardless of application, it is our vision to provide you with the product you need, ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual, you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly, as per the experiment.
- In Advance Fiber Optic Experimental Modules, signals are to be monitored with a Digital Storage Oscilloscope, or with the help of indicators as explained in the manual. The scope input channel should be A.C. coupled, unless otherwise indicated. For observation of signals on oscilloscope use X10 (Attenuation Probe).
- **Laser Safety**
 - It is imperative that you read and understand the following safety
 - Information before attempting any of the experiments from this manual.
 - If you have inquiries, please consult your lecturer or demonstrator.
- **Operational Hazard-Semiconductor Laser Diode**

The laser diode source emits radiation at 1310, which is Invisible to the human eye and may cause **eye damage** if the output beam is viewed directly. While using, never stare at the optical output port of the Laser Diode, when its dust cap is removed. Always leave the dust cap for the fiber patch cord in position until ready to make a connection

 - And never stare at the end of a fiber patch cord connected to the Laser
 - Diode output port or point the fiber end at yourself or anyone else in your
 - Vicinity before making a connection. Never use external optics (i.e.
 - Collimating lenses) to view the Laser Diode emission at a close range.
 - The output power of the semiconductor Laser Diode is less than 1mW which is in line with all the applicable standards of DHMS regulations and falls within the international safety Class3B.
- This kit must be employed only for the use for which it has been conceived, i.e. as an educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions's sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions's judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

Source 1

Type	: Laser
Central wavelength	: 1310nm
Spectral width	: 1nm
Output power	: 1.5mW
Threshold Current	: 15mA Max
Forward Current	: 85mA Max
Feedback	: Built-in InGaAs monitor detector
Receptacle Housing	: SC

Source 2

Type	: LED red visible
Central wavelength	: 660nm
Spectral Width	: 25nm
Output Power	: 200 μ W@ $I_f = 10$ mA
Forward Current	: 50mA Max
Receptacle Housing	: "Connector-less" Style package

Detector 1

Type	: InGaAs PIN photo diode
Spectral Bandwidth	: 1250nm ~ 1600nm
Responsivity	: 0.9 A/W @ 10 μ W of 1310nm
Bandwidth	: 1.5 GHz
Return loss	: 14dB
Reverse voltage, VR	: 30V (Maximum)
Reverse current, IR	: 500 μ A Max.

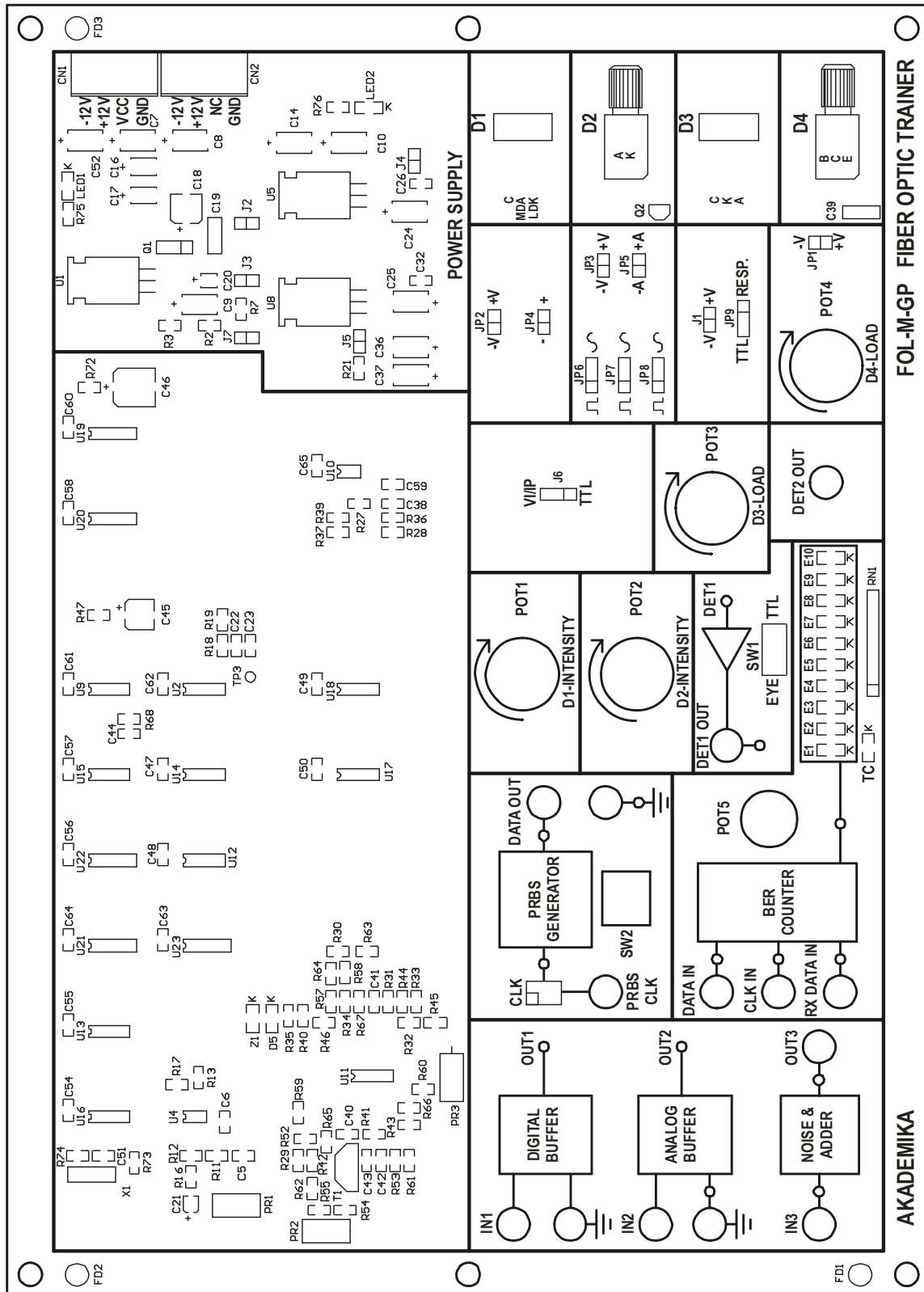
Detector 2

Type	: Silicon PIN photo transistor
Spectral bandwidth	: 400nm ~ 1100nm
Max. Photo sensitivity λ	: 850nm
Responsivity	: 0.8 mA / 10 μ W VCE = 5V
Rise and fall time	: 20ms

Fiber cable cable:

Type	: Glass fiber single & multimode
Core diameter	: 62.5 and 9 μ m
Clad diameter	: 125 μ m
Primary coating diameter	: 250 μ m
Total outer diameter	: 2.2mm
Number of fibers	: 02
Length of fiber	: 01 meter each
Type	: plastic optical cable, step index, multimode
Core refractive index-n1	: 1.492
Clad refractive index-n2	: 1.406
Numerical aperture	: 0.50
Acceptance angle	: 60°
Fiber diameter	: 1000microns
Outer diameter	: 2.2mm

Functional Block Diagram



INTRODUCTION

Fiber optics technology is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All giant telephone companies like AT & T, sprints have saturated the airways & print media with advertisements heralding this & new technology. Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications.

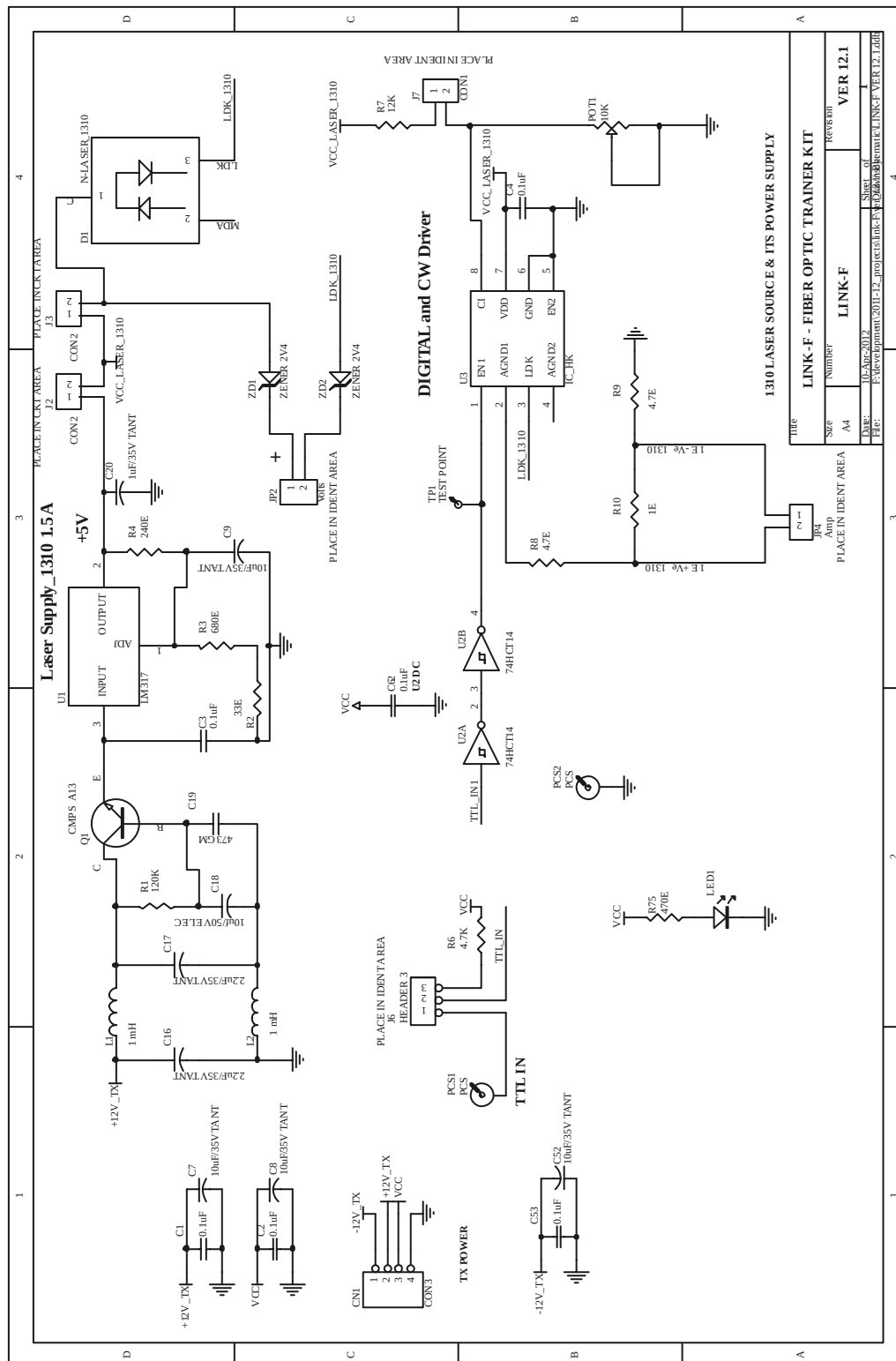
And their enthusiasm is valid. Today medical fiber optics systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

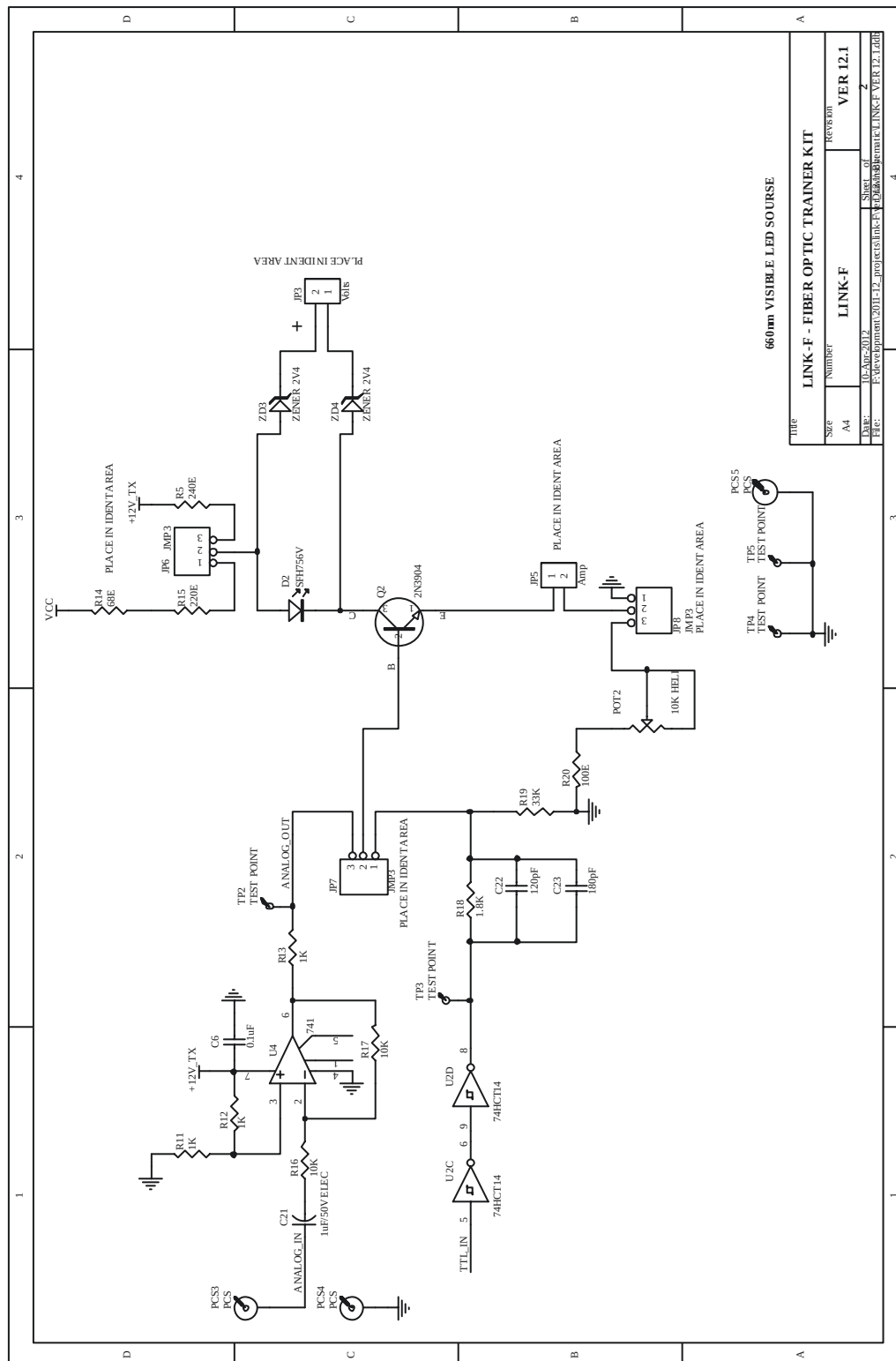
Very few technologists ever come close to the fantastic growth rate that market analyst love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influence of fiber optics will pervade our homes workspaces & recreational facilities. Your television reception will take on startling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronics newspapers become feasible. Two-way fiber optics will allow you to “attend college” in the comfort of your home.

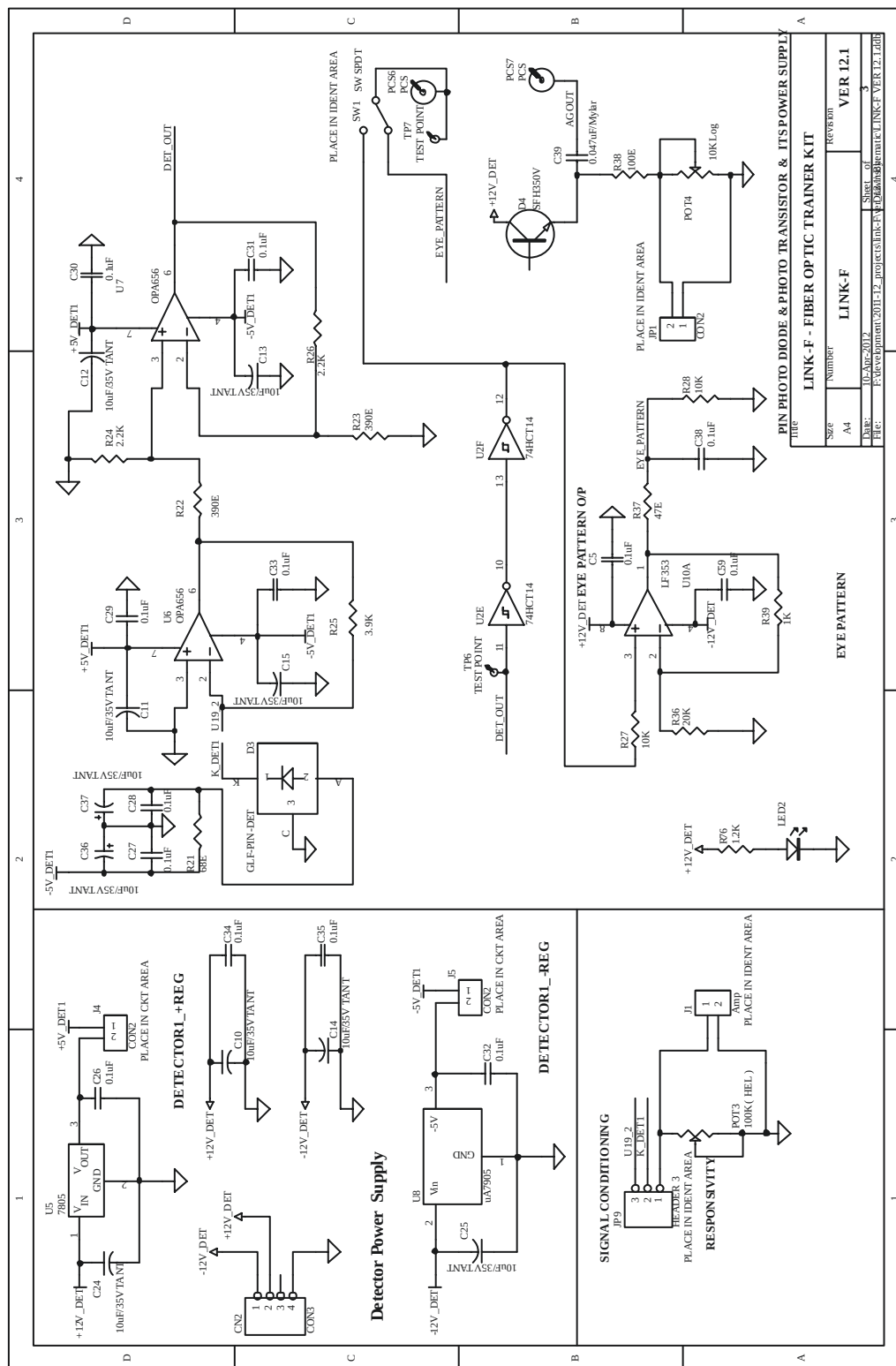
Anyone who takes an interest in fiber optics & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few experts exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons existing with the prospect of making new discoveries finding new application for the technology.

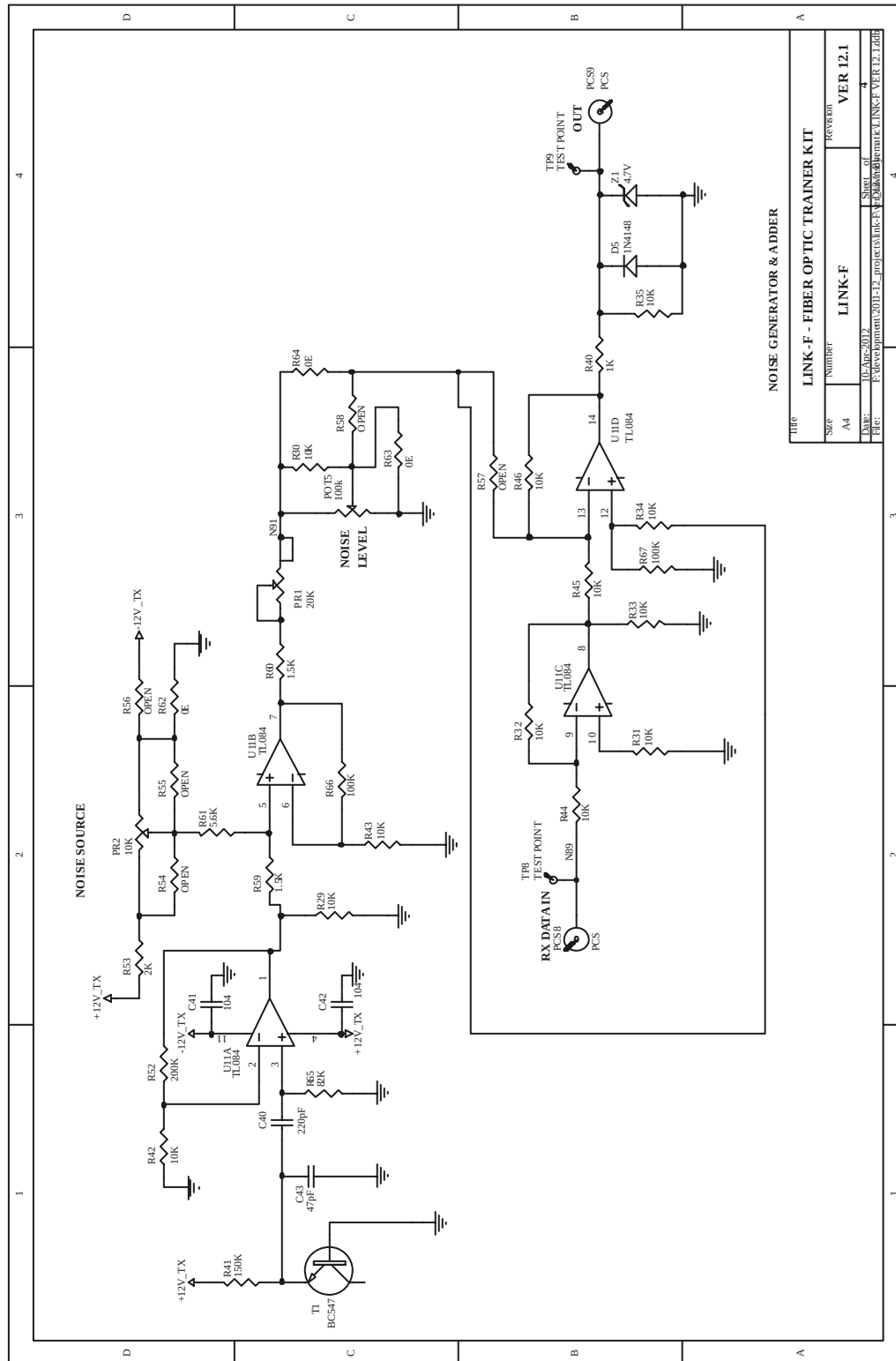
This kit is an introduction to fiber optics communication technology for instructors, students & hobbyist. Welcome to fascinating & expanding world of fiber optics. We hope that you find the kit challenging, stimulating & -yes- even fun.

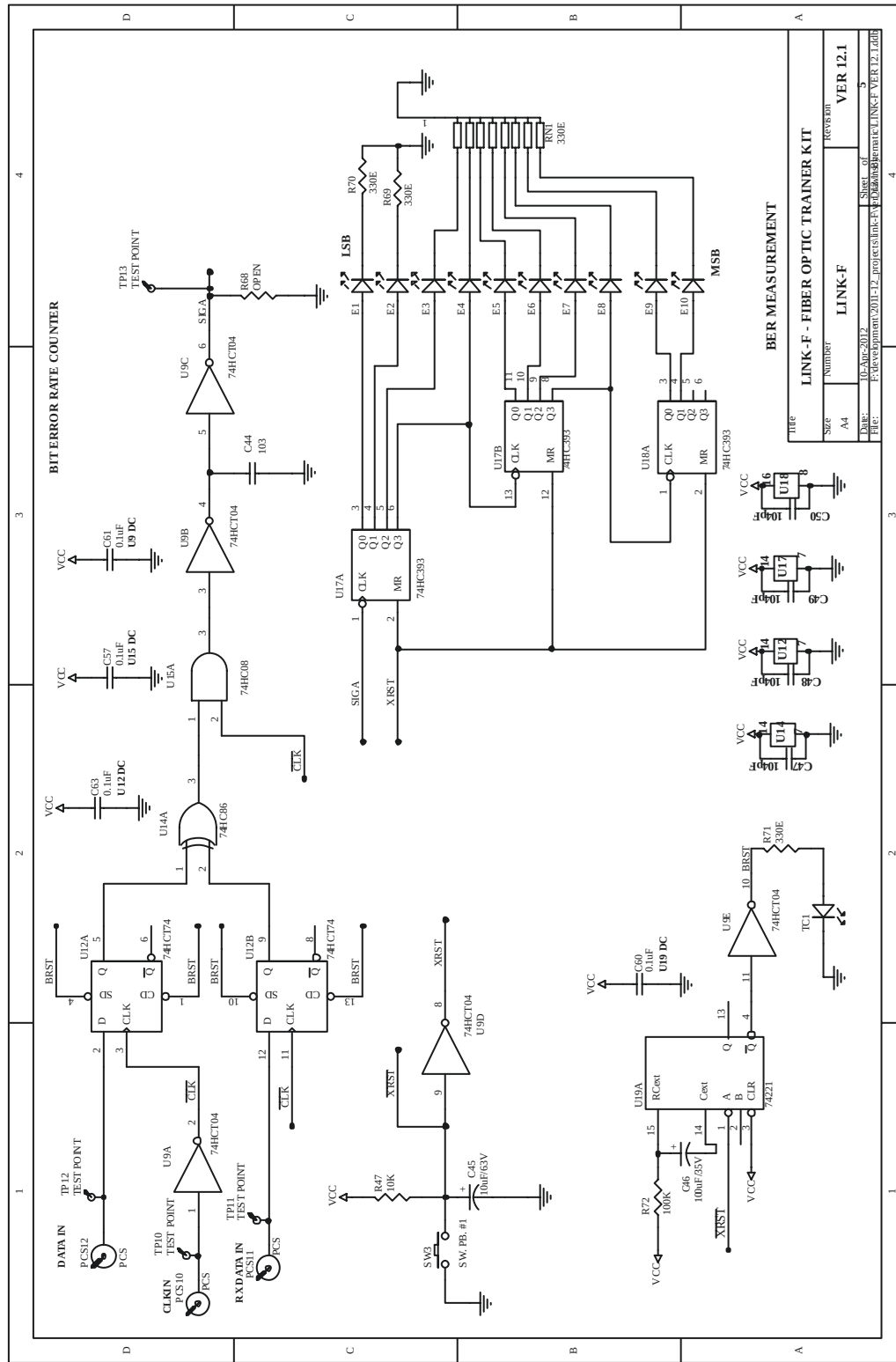
FOL-M-GP circuit diagram

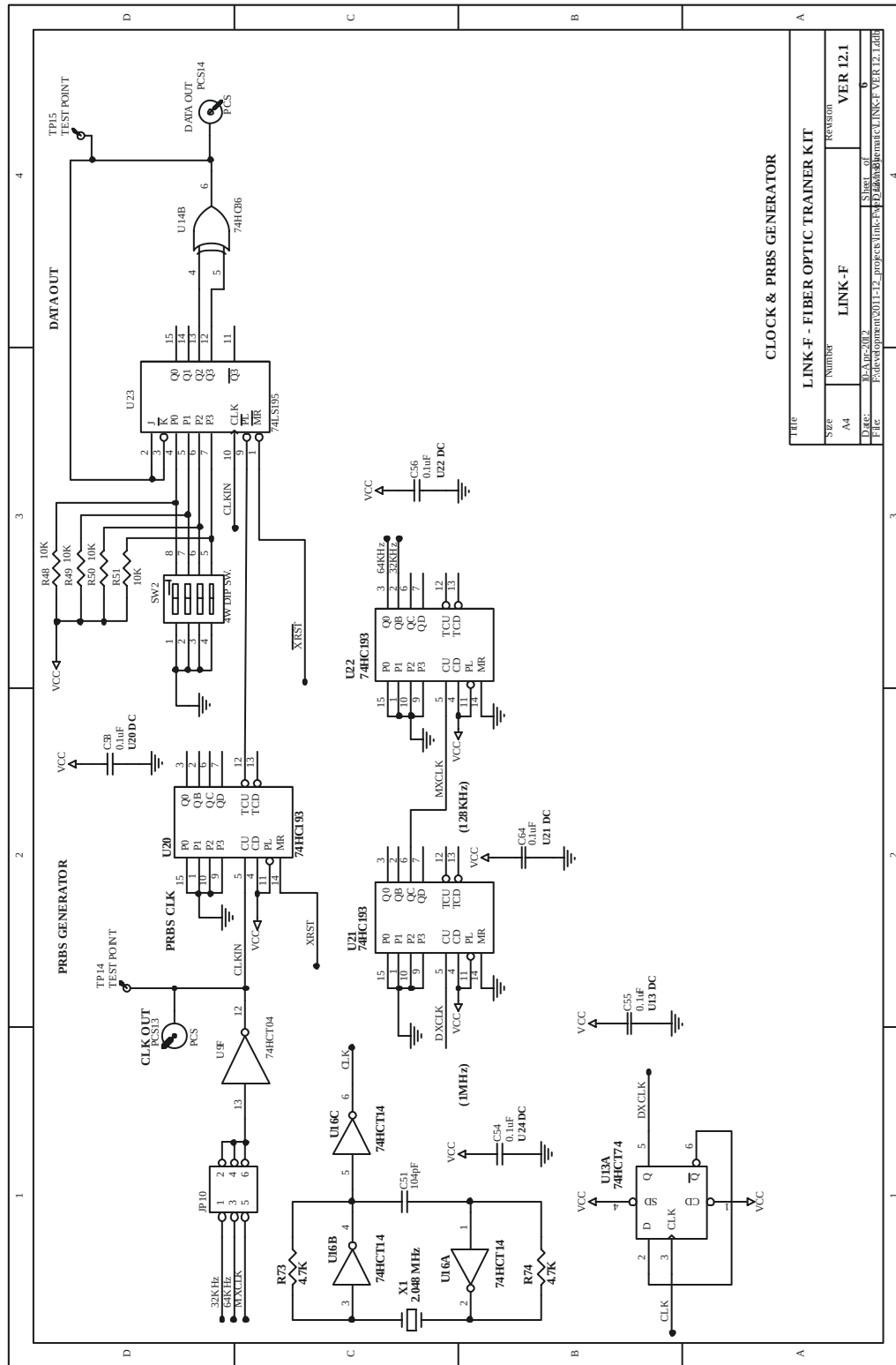












ABOUT FIBER OPTICS

Before fiber optics came along the primary means of real time, data communication was electrical in nature. It was accomplished using copper wire or by transmitting electromagnetic waves through free space. Fiber optics changed that by providing an alternative means of sending information over significant distances over light energy. Although, Fiber Optics was a very controversial technology initially, today, it is considered as reliable & and cost effective.

High bandwidth properties of fiber cables can be used over greater distances because of the Low Loss as compared to copper cables. In data networks, this can be as much as 2Km without the use of repeaters. Their light weight and small size also make them ideal for applications, where running copper cables would be impractical. By using multiplexers, one fiber can replace hundreds of copper cables. This is pretty impressive for a tiny glass filament, but the real benefits in the data industry are its immunity to Electro Magnetic Interference (EMI), and the fact that glass is not an electrical conductor. Fiber is non-conductive, therefore, it can be used wherever electrical isolation is required. For instance, it can be used between buildings where copper cables would require cross bonding to eliminate differences in earth potentials. Also, Fibers do not pose any threat in dangerous environments such as chemical plants, where a spark could trigger an explosion. Last but not least is the security aspect. It is very, very difficult to tap into a fiber cable to read the data signals

You will learn soon that fiber optic communication is indeed a simple technology, closely related to electronics. In fact, it was research in electronics that set the stage for fiber optics to develop into the communication giant that it is today.

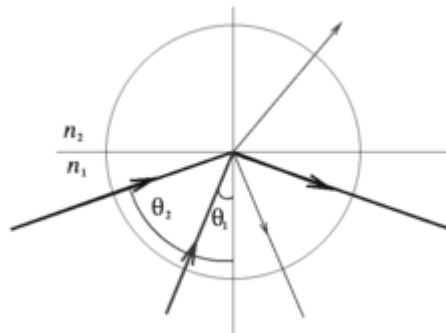
How Optical Fiber Works?

The principle of operation of optical fiber lies in the behavior of light. Light travels in straight line through most optical materials, but that's not necessarily the case at the junction (interface) of two materials of different refractive indices. Think of a fiber cable in terms of a very long cardboard roll (from the inside roll of paper towel) that is coated with a mirror. If you shine a flashlight in one you can see light at the far end - even if you bend the roll around a corner.

Light pulses move easily down the fiber-optic line because of a principle known as the total internal reflection. "This principle of the total internal reflection states that when the angle of incidence exceeds a critical value, light cannot get out of the glass; instead, the light bounces back in as shown in figure. When this principle is applied to the construction of the fiber-optic strand, it is possible to transmit information through fiber lines in the form of light pulses.

Total internal reflection is an optical phenomenon that occurs when light strikes a medium boundary at a steep angle. If the refractive index is lesser on the other side of the boundary, no light can pass through, so all of the light gets reflected effectively. The critical angle is the angle of incidence above which the total internal reflection occurs.

When light crosses a boundary between materials with different refractive indices, the light beam will be partially refracted at the boundary surface and partially it will get reflected. However, if the angle of incidence is shallower (closer to the boundary) than the critical angle (the angle of incidence where light is refracted so that it travels along the boundary), then the light will stop crossing the boundary altogether, instead it will reflect back totally internally. This can only occur where light travels from a medium with a higher refractive index to the one with a lower refractive index. For example, it will occur when light passes from glass to air, but not when light would pass from air to glass.

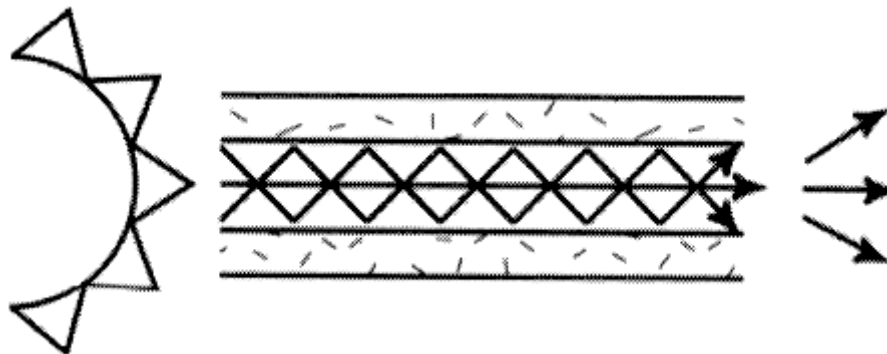


Total internal reflection

Types of Optical Fibers

The three types of fiber optic cables commonly used are as follows:

- Single Mode
- Multimode
- Plastic Optical Fiber (POF)



Fiber optic cable functions as a 'light guide', guiding the light introduced at one end of the cable to the other end. The light source can either be a light-emitting diode (LED) or a laser.

The light source is pulsed on and off, and a light-sensitive receiver on the other end of the cable converts the pulses back into the digital ones and zeros, of the original signal.

Even laser light shining through a fiber optic cable is subject to loss of strength, primarily through dispersion and scattering of the light, within the cable itself. The faster the laser fluctuates, the greater the risk of dispersion. Light strengtheners, called repeaters, may be necessary to refresh the signal in certain applications.

While fiber optic cable itself has become cheaper over time - an equivalent length of copper cable costs lesser per foot, but not in capacity. Fiber optic cable connectors and the equipment required to install them are more expensive than their copper counterparts.

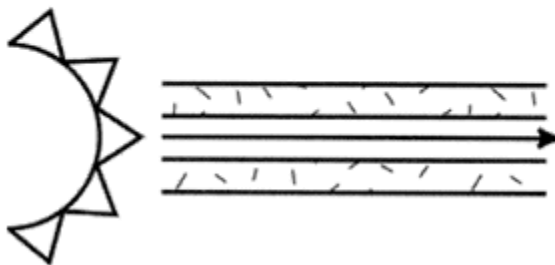
Single Mode Cable

It is a single strand of glass fiber with a diameter of 8.3 to 10 microns that has one mode of transmission. Single Mode Fiber with a relatively narrow diameter, through which only one mode will propagate typically 1310nm or 1550nm, carries higher bandwidth than multimode fiber, but requires a light source with a narrow spectral width.

Single-mode fiber gives you a higher transmission rate and up to 50 times more distance than multimode, but it also costs more. Single-mode fiber has a much smaller core than multimode. The small core and single light-wave virtually eliminate any distortion that could result from overlapping light pulses, providing the least signal attenuation and the highest transmission speeds of any fiber cable type.

Single-mode optical fiber is an optical fiber in which only the lowest order bound mode can propagate at the wavelength of interest, typically 1300nm to 1320nm.

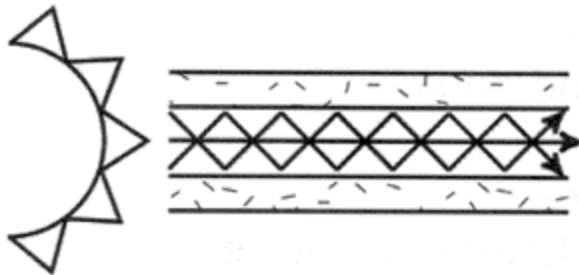
“Single mode fiber”
single path through the fiber



Multimode Cable

It is made of glass fibers, with common diameters in the 50-to-100 micron range for the light carry component (the most common size is $62.5\mu\text{m}$). Multimode fiber gives you high bandwidth at high speeds over medium distances. Light waves are dispersed into numerous paths, or modes, as they travel through the cable's core, typically 850nm or 1300nm. Typical multimode fiber core diameters are 50, 62.5, and 100 micrometers. However, in long cable runs (greater than 3000 feet [914.4 meters]), multiple paths of light can cause signal distortion at the receiving end, resulting in an unclear and incomplete data transmission.

“Multimode fiber”
multiple paths through the fiber



Plastic Optical Fiber (POF)

It is a newer plastic-based cable which promises performance, similar to glass cable on very short runs, but at a lower cost.

Single Mode V/S Multimode

A fiber can support one or several (more than one) propagation modes, the intensity distributions of which are located at or immediately around the fiber core, although some of the intensity may propagate within the fiber cladding. Other modes are not restricted to the core region and are called cladding modes. The power in these are usually lost after some distance of propagation, but it can in some cases propagate over longer distances. Outside the cladding, there is typically a protective polymer coating, which gives the fiber improved mechanical strength and protection against moisture and also determines the losses for cladding modes.

Some important differences between single-mode and multimode fibers are mentioned below:

- Single-mode fibers usually have a relatively small core (with a diameter of only a few micrometers) and can guide only a single spatial mode

- (disregarding the fact that there are two different polarization directions), the profile of which in most cases has roughly a Gaussian shape. Changing the launch conditions only affects the launched power, while the spatial distribution of the light exiting the fiber is fixed. Efficiently launching light into a single-mode fiber usually requires a laser source with good beam quality and precise alignment of the focusing optics, in order to achieve mode matching. Also, there are large mode area fibers with single-mode guidance, where the alignment tolerances are lower in terms of position, but higher in terms of angle (which is not considered as a big problem).
- Multimode fibers usually have a larger core and / or a larger index difference between core and cladding, so that they support multiple modes with different intensity distributions (see the figure below). In this case, the spatial profile of light exiting the fiber core may depend on the launch conditions, which determine the distribution of power among the spatial modes.

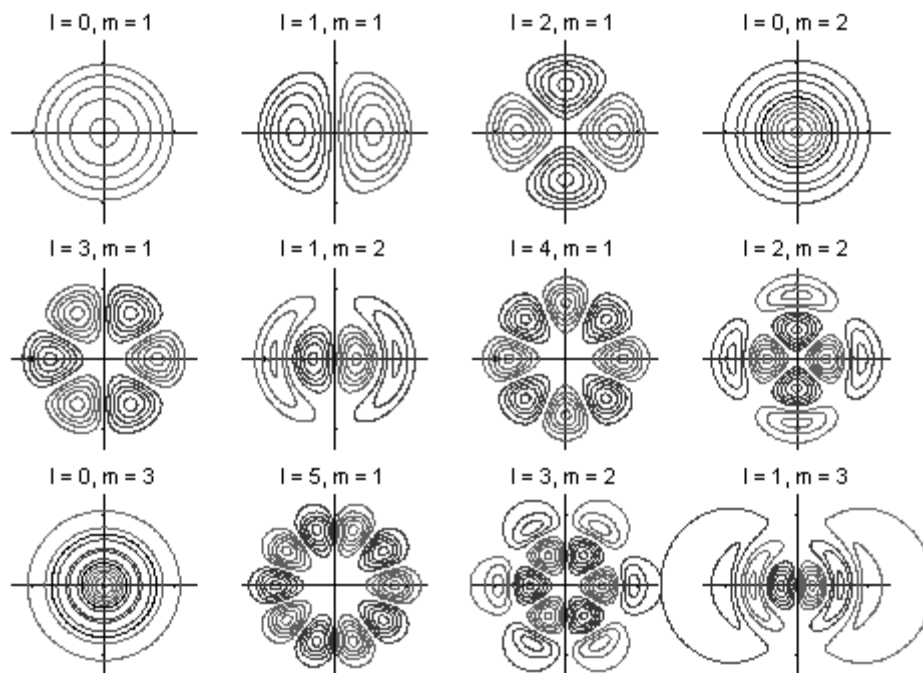


Fig.: Modes of a fiber with a top-hat refractive index profile.

The lowest-order mode ($l = 1, m = 0$, called LP_{01} mode) has an intensity profile which is similar to that of a Gaussian beam. Note that for a given wavelength, the number of fiber modes is finite; e.g., the figure shows all modes for the given configuration. In general, light launched into a multimode fiber will excite a superposition of different modes, which can have a rather complicated shape.

Long-range optical fiber communications systems usually use single-mode fibers, because the different group velocities of different modes would mess up the data at high data rates. However, for shorter distances, multimode fibers are more convenient as the demand on light sources and component alignment are lower. Therefore, local area networks (LANs), except those for highest bandwidth, normally use multimode fiber.

Single-mode fibers are also normally used for fiber lasers and amplifiers. Example: Multimode fibers are often used to transport light from a laser source to the place where it is required, particularly when the light source has a poor beam quality and / or the high optical power requires a large mode area.

Different modes of a fiber can be coupled via various effects. Example: By bending or often by irregularities in the refractive index profile. These may be unwanted or purposely introduced. E.g. As fiber Bragg gratings. Waveguides theory shows that an important factor for the coupling between different fiber modes is the difference of their wave numbers, which for efficient coupling has to match the wave number of a coupling disturbance.

Advantages of Fiber Optics Over Conventional Copper Cables

- **Less expensive** - Several miles of optical cable can be made at cheaper rates compared to equivalent lengths of copper wires. This saves your service providers (cable TV, Internet) and your money too.
- **Thinner** - Optical fibers can be drawn to smaller diameters than copper wire.
- **Higher carrying capacity** - Optical fibers are thinner than copper wires, therefore, more fibers can be bundled within a given diameter cable than the copper wires. This allows more phone lines to go over the same cable or more channels to pass through the cable into your cable TV box.
- **Less signal degradation** - The loss of signal in optical fiber is lesser than that observed in copper wires.
- **Light signals** - Unlike electrical signals in copper wires, light signals from one fiber does not interfere with those of the other fibers in the same cable. This means clearer phone conversations or TV reception.
- **Low power** - Signals in optical fibers degrade less; therefore, lower-power transmitters can be used instead of the high-voltage electrical transmitters, as required for copper wires. Again, this saves your provider and your money.
- **Digital signals** - Optical fibers are ideally suited for carrying digital information, which is especially useful in computer networks.
- **Non-flammable** - No electricity is passed through optical fibers; therefore, there is no fire hazard.
- **Lightweight** - An optical cable weighs lesser than a comparable copper wire cable. Also, Fiber-optic cables take very little space in the ground.

- **Flexible** - Fiber optics are very flexible and can transmit and receive light effectively; therefore, they are used in many flexible digital cameras for the various purposes. Some of the examples are mentioned below:
 - **Medical imaging** – Used in bronchoscopes, endoscopes and laparoscopes.
 - **Mechanical imaging** – To inspect mechanical welds along pipes and engines (airplanes, rockets, space shuttles, cars).
 - **Plumbing** - To inspect sewer lines.

With all the above properties of optical fiber, it is becoming very important in sensing fields too. Most of the properties of optical fiber are useful in sensors. Further, optical fiber sensors offer extreme sensitivity & can be used to sense almost any physical parameters like temperature, velocity, current, voltage etc.

EXPERIMENT NO.1

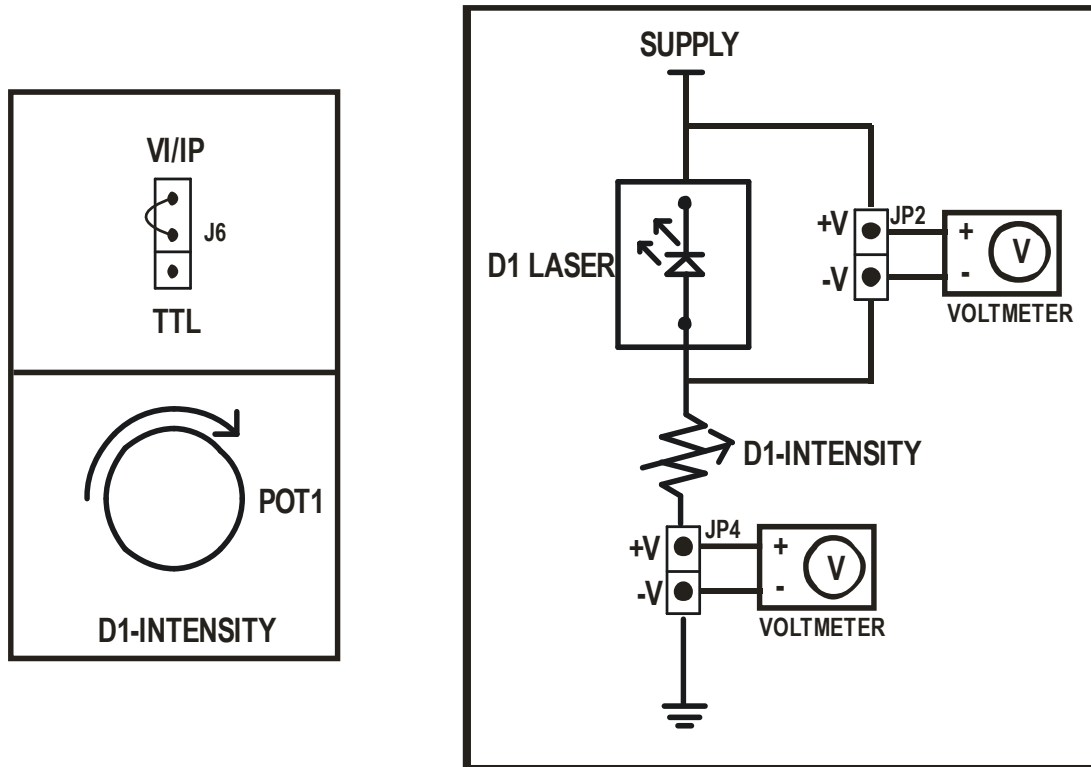


FIG 1A EXPERIMENTAL SETUP FOR VI CHARACTERISTIC OF LASER DIODE.

EXPERIMENT NO. 1

NAME

- A) Plot the V-I characteristics for 1310nm Laser source for the CW
- B) Plot the V-I characteristics for 660nm LED source for the CW
- C) Plot the I-P characteristics for 1310nm Laser source for the CW

OBJECTIVE

To study and plot the V-I characteristics for 1310nm Laser source for the CW

EQUIPMENTS

- FOL-M-GP
- Power supply
- Voltmeters/ DMM
- Optical power meter
- 1 meter single mode fiber Patch chord
- Jumper to Crocodile Links

PROCEDURE

A) V-I characteristics for 1310nm Laser source

- Refer to fig 1a and carry out the connections
- Make sure that the power supply of FOL-M-GP is off.
- Keep Intensity Control pot POT1 fully anticlockwise.
- Keep jumper J6 to VI/IP.
- Connect voltmeter between jumper JP2 with respective polarity, which indicates voltage across the laser.
- Connect voltmeter between jumper JP4 with respective polarity, which indicates current through LASER in terms of voltage.

Note: by referring schematic given above, for current measurement, we are actually measuring the voltage across 1E resistor and by using OHMS law $V=IR$, we can directly get current flowing through laser.

E.g. if the voltmeter reads 50mV then it corresponds to 50 mA current.

- Turn the Intensity Control pot slightly in the clockwise direction

- Record the current in mA and the corresponding voltage in volts.
- Repeat the procedure for current readings up to the 45mA.
- Plot the graph for Current (mA) vs Voltage (V).
- Find out the Threshold Current (I_{th}) in mA from the I-V graph.

OBSERVATION:

Laser Sr No:

[illegible]

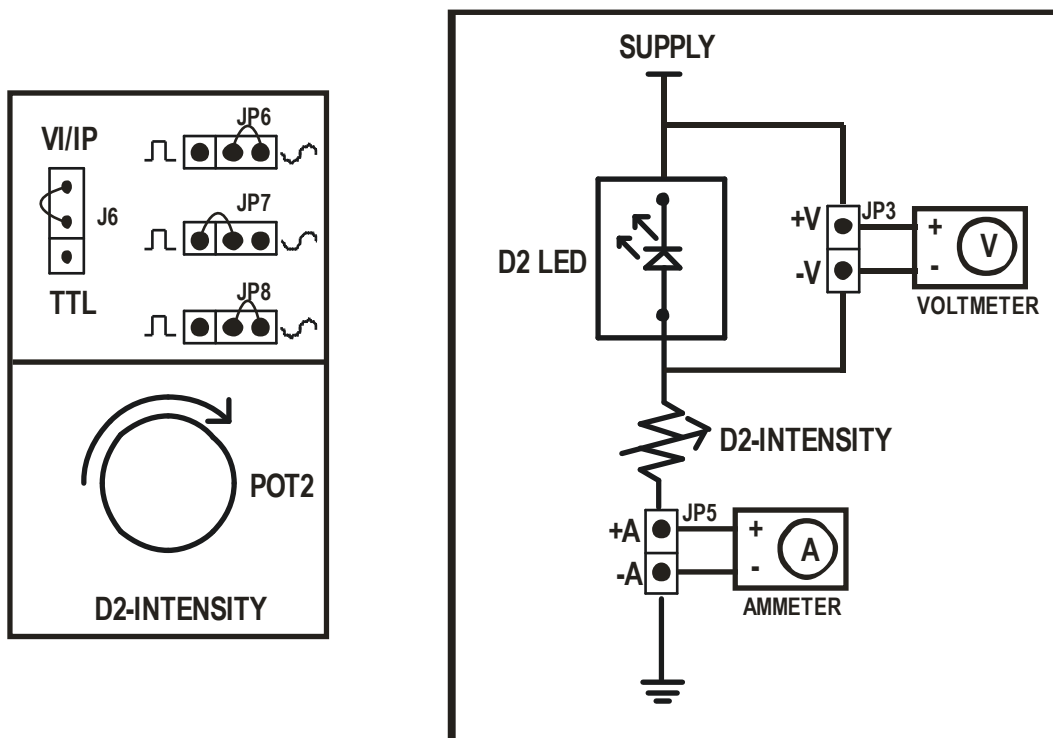


FIG 1B EXPERIMENTAL SETUP FOR VI CHARACTERISTIC OF 660nm LED.

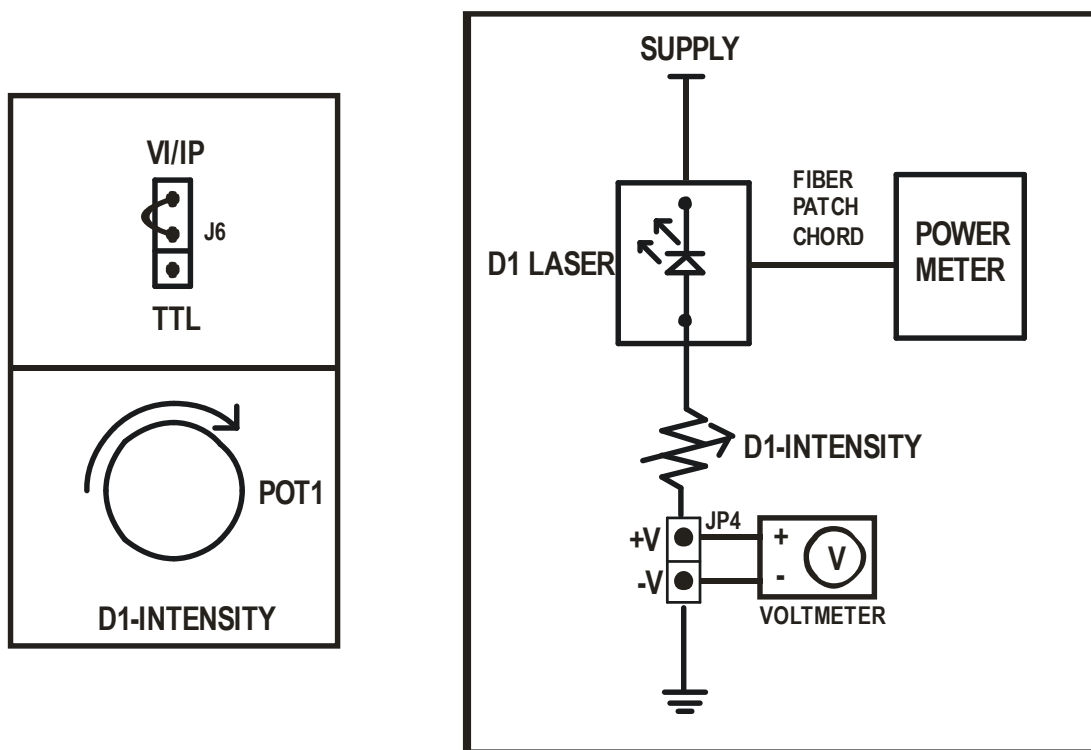


FIG 1C EXPERIMENTAL SETUP FOR I-P CHARACTERISTIC OF LASER DIODE.

C) I-P characteristics for 1310nm Laser source

- Refer to fig 1C and carry out the connections
- Make sure that the power supply of FOL-M-GP is off.
- Keep Intensity Control pot POT1 fully anticlockwise.
- Keep jumper J6 to VI/IP.
- Connect voltmeter between jumper JP2 with respective polarity, which indicates voltage across the laser.
- Connect voltmeter between jumper JP4 with respective polarity, which indicates current through LASER in terms of voltage.

Note: by referring schematic given above, for current measurement, we are actually measuring the voltage across 1E resistor and by using OHMs law $V=IR$, we can directly get current flowing through laser.

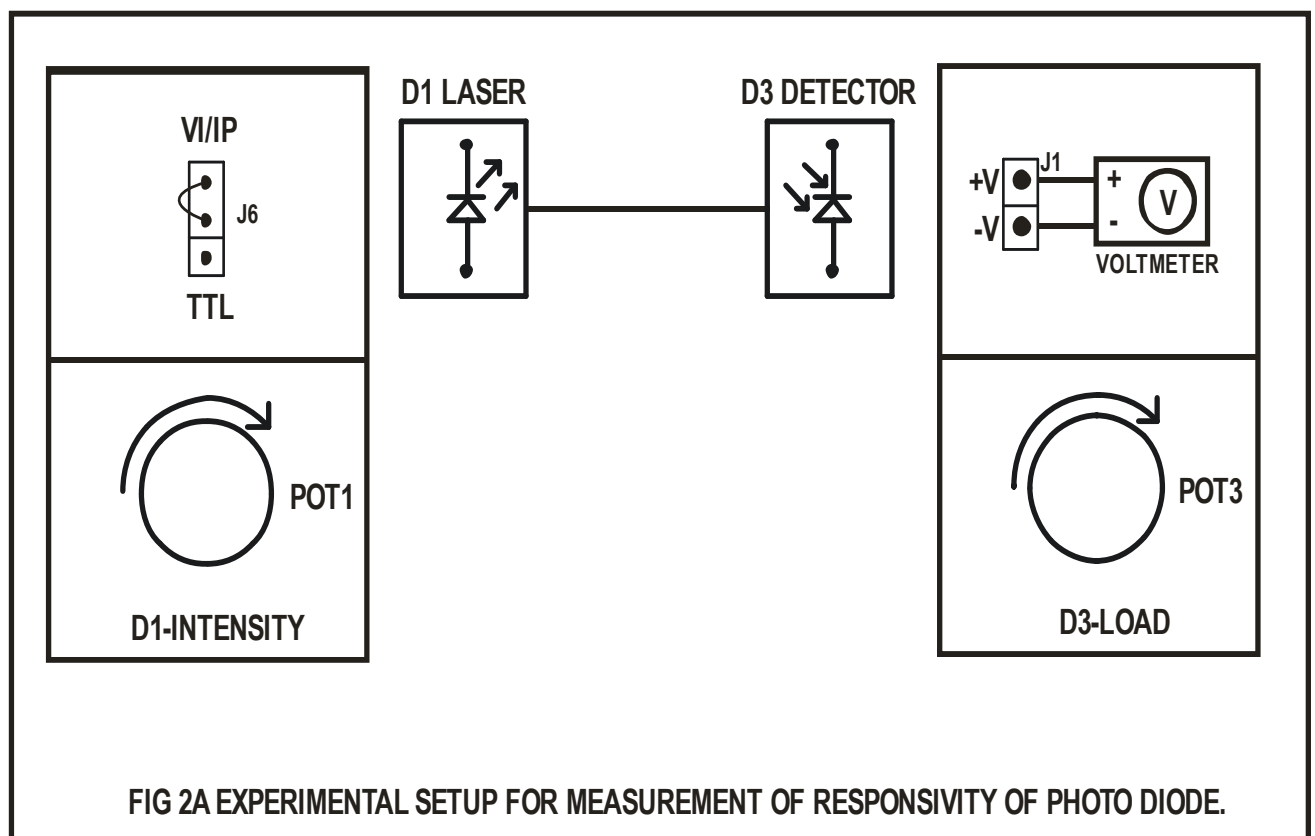
E.g. if the voltmeter reads 50mV then it corresponds to 50 mA current.

- Connect Laser output to optical power meter using 1 meter single mode patch chord provided. Keep wavelength setting of power meter to 1310nm.
- Turn the Intensity Control pot slightly in the clockwise direction
- Record the current in mA and the corresponding power in Watts.
- Repeat the procedure for current readings up to the 45mA.
- Plot the graph for Current (mA) VS Power (W).
- Find out the Threshold Current (I_{th}) in mA from the I-P graph after which laser enter into lasing action.

Laser Sr No:

[illegible]

EXPERIMENT NO.2



EXPERIMENT NO. 2

NAME

To study and measure the Responsivity and frequency response of detector

OBJECTIVE

- A) Measure the Responsivity of the pin photo diode
- B) Plot the frequency response of pin photo transistor with different values of load.

EQUIPMENTS

- FOL-M-GP
- Power supply
- Voltmeters/ DMM
- Optical power meter
- 1 meter single mode fiber Patch chord
- 1 meter plastic fiber
- Connecting Links

PROCEDURE

A) Responsivity of PIN photo diode

1. To perform this experiment a stable reference source of light at about 1310nm is required.
2. The 1310 nm laser current can be set by a potentiometer POT1. By referring experimental procedure A) connect the voltmeters to observe the voltage across and current flowing through Laser.
3. Keep the jumper JP9 to RESP. position
4. Connect voltmeter between J1 with proper polarity.
5. Connect power supply to the FOL-M-GP trainer and switch it ON.
6. Connect Laser to the optical power meter using the fiber patch chord provided and adjust the optical power to 10 μ W. while adjusting this make sure that laser must be in lasing mode(use 10dB attenuator to reduce the power to 10 μ W.)
7. Note down the exact optical power for reference. Ensure that fiber optic power meter is set for operation of 1310nm, by pressing the λ key.
8. **The optical power level set is now the reference power for rest of the exercise and should not be altered again. Make sure also that the fiber is not under stress or tightly wound when taking the reference reading as this could make the reference setting invalid.**

9. Connect laser source which is set at 10 μW to D3 using patch chord provided.
10. The current in the Photo detector can be measured by connecting an voltmeter across load resistor POT3 using J1.
11. Note down the voltmeter reading (which is generally in mV range). Convert this voltage reading to current using ohms law. In our case $R = 100\text{K}\Omega$.
12. Now set the o/p power to 20 μW , 30 μW , 40 μW & 50 μW and repeat the procedure from 9 to 11
13. Verify the Responsivity of the photo diode with datasheet.

OBSERVATIONS:

Case I) I/p power 10 μW @1310 laser

Voltage across POT3 = _____ V

By ohms law current $I =$ _____ I

Responsivity of detector @10 $\mu\text{W} =$ _____ A/W

Case II) I/p power 20 μW @1310 laser

Voltage across POT3 = _____ V

By ohms law current $I =$ _____ I

Responsivity of detector @10 $\mu\text{W} =$ _____ A/W

Case III) I/p power 30 μW @1310 laser

Voltage across POT3 = _____ V

By ohms law current $I =$ _____ I

Responsivity of detector @10 $\mu\text{W} =$ _____ A/W

Case IV) I/p power 40 μW @1310 laser

Voltage across POT3 = _____ V

By ohms law current $I =$ _____ I

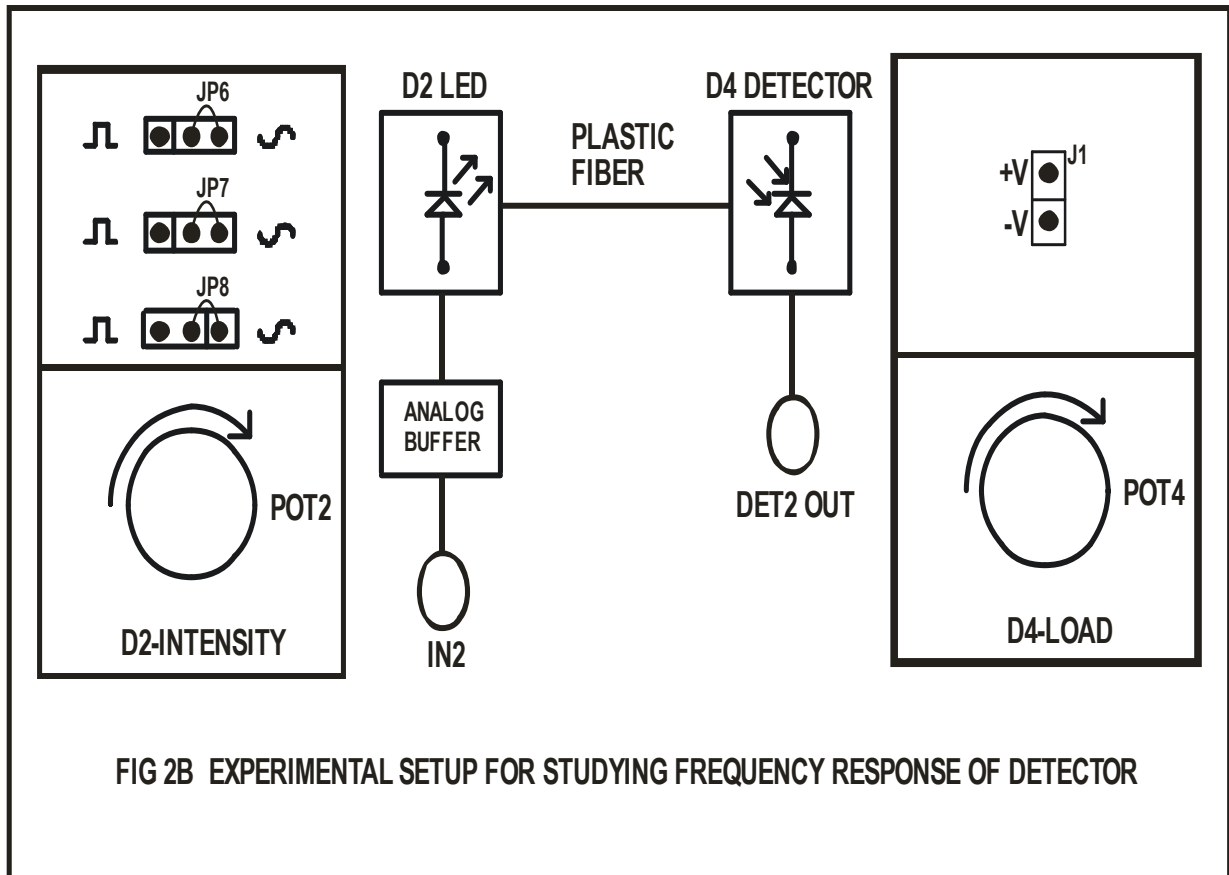
Responsivity of detector @10 $\mu\text{W} =$ _____ A/W

Case V) I/p power 50 μW @1310 laser

Voltage across POT3 = _____ V

By ohms law current $I = \frac{V}{R}$

Responsivity of detector @ $10 \mu\text{W} = \frac{I}{P}$ A/W



B) Frequency response of PIN photo transistor.

1. Do the experimental setup as shown in Fig. 2B.
2. APPLY 1 KHz, 1Vp-p sine wave from function generator to IN2 post.
3. Short jumpers JP6, JP7 and JP8 to analog position.
4. Connect Optical O/P of LED D2 to photo transistor D4 using 1 meter plastic fiber patch chord provided.
5. Connect DMM between J1 with proper polarity and keep it on resistance / Ohm mode. Now set the resistance of Load D4 to 500E and remove the DMM from J1
6. Connect power supply to the FOL-M-GP trainer and switch it ON.
7. Now observe the output at DET2 OUT post, and set it to 1Vp-p using intensity control POT2.
8. Now vary the frequency from function generator and find out the bandwidth at same setting. Note down the readings.
9. Switch of the power supply and repeat the procedure from 5 to 8 for different load values.
10. By observing the reading we conclude that the response of detector depends on load.

OBSERVATIONS:

Case I) Resistance of load POT4 = 500E

Observed bandwidth = _____ KHz.

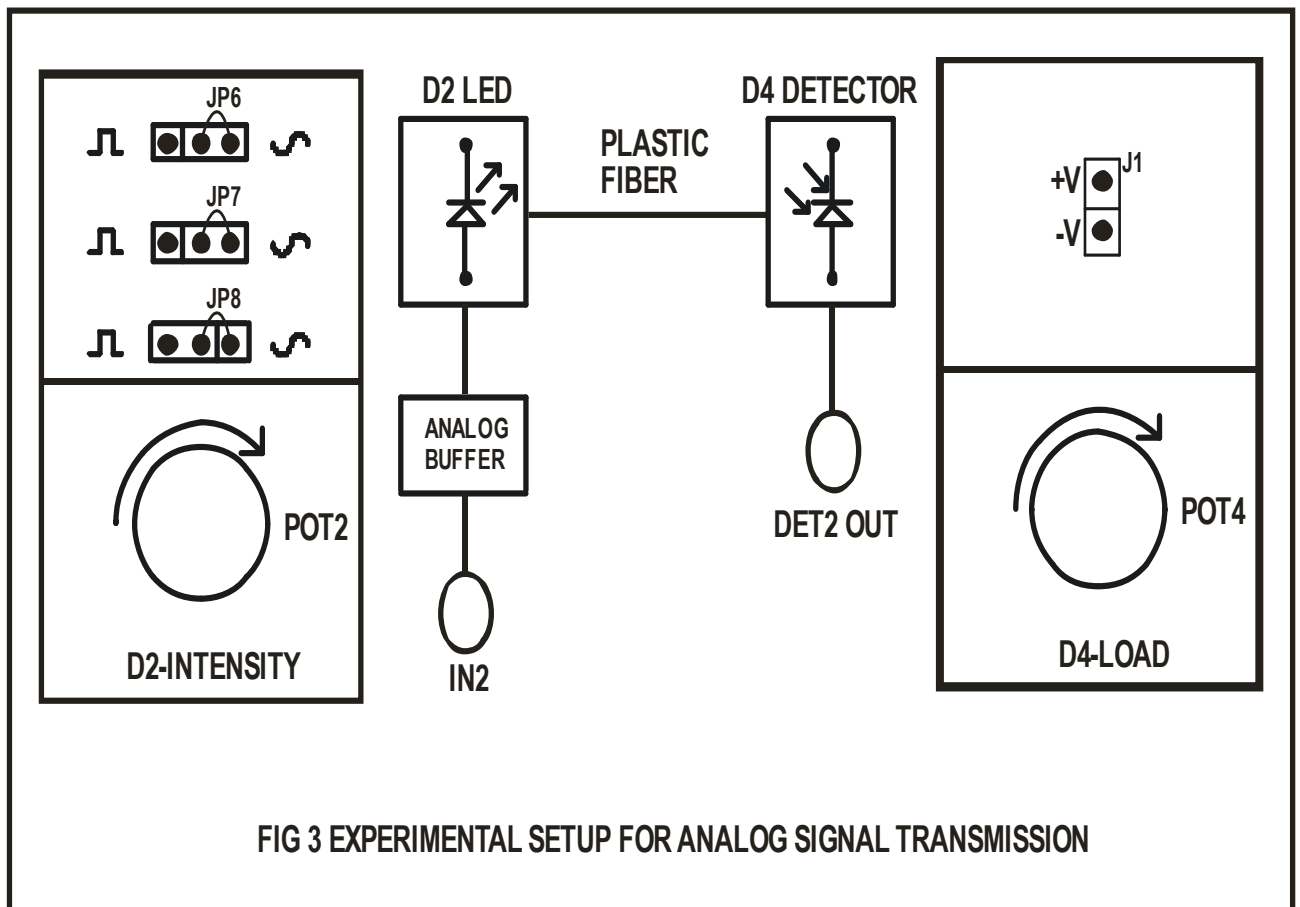
Case II) Resistance of load POT4 = 1KE

Observed bandwidth = _____ KHz.

Case III) Resistance of load POT4 = 1.5KE

Observed bandwidth = _____ KHz.

EXPERIMENT NO.3



EXPERIMENT NO.3

NAME

Analog Signal transmission for 660nm LED Source.

OBJECTIVE

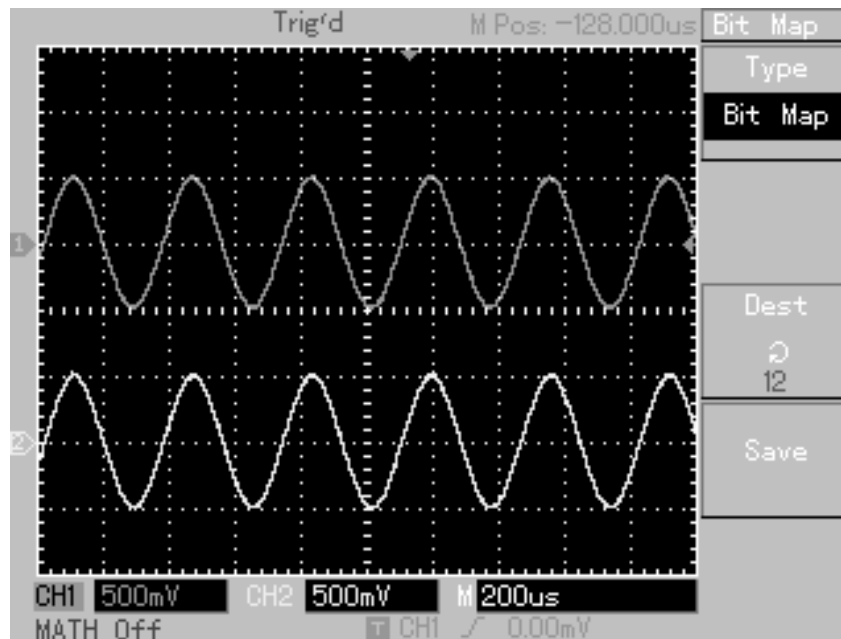
To study the Analog Signal transmission for 660nm LED Source.

EQUIPMENTS

- FOL-M-GP
- Power supply
- 1 meter plastic fiber patch chord
- Function generator
- DSO
- Connecting Links

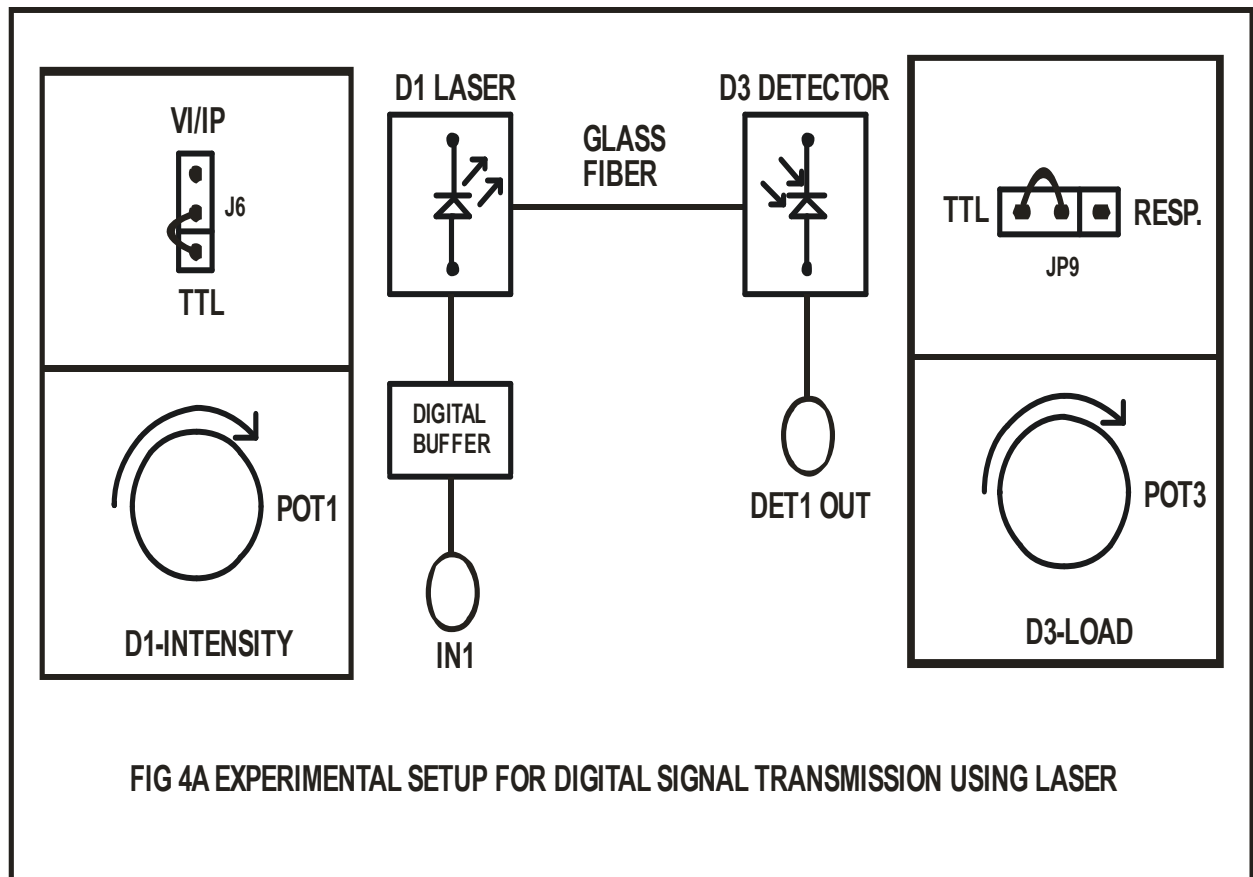
PROCEDURE

- Refer to fig 3 and carry out the connections
- Keep Intensity Control pot POT2 fully anticlockwise.
- Keep jumper JP6, JP7 and JP8 to analog mode.
- Make sure that the power supply of FOL-M-GP is off.
- Apply 1Vp-p sine wave of frequency 1 KHz from function generator to the IN2 post of analog buffer.
- Connect Optical O/P of LED D2 to photo transistor D4 using 1 meter plastic fiber patch chord provided.
- Vary the Intensity control Pot POT2 and observe the detected signal at DET2 OUT post.
- Go on increasing the Function Generator sine wave frequency in steps of 1 KHz and observe the Detected Signal.
- Note down the Frequency for the Function Generator sine wave at which the Detected Signal O/P. to get the undistorted detected signal adjust load using POT 4.

OBSERVATIONS:

CH1 : input signal from Function generator
CH2: detected signal through fiber optic link

EXPERIMENT NO.4



EXPERIMENT NO.4

NAME

Digital Signal transmission using 1310nm LASER and 660nm LED Source.

OBJECTIVE

- A) To study the digital Signal transmission using 1310nm LASER
- B) To study the digital Signal transmission using 660nm LED Source.

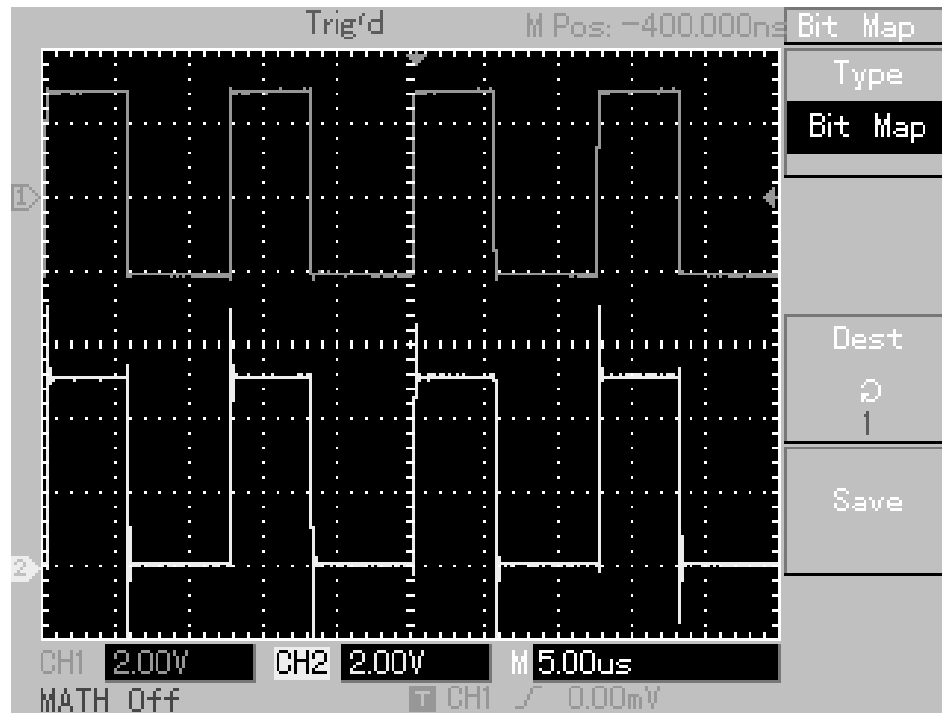
EQUIPMENTS

- FOL-M-GP
- Power supply
- 1 meter plastic fiber
- 1 meter single mode glass fiber patch chord
- Function generator
- DSO
- Connecting Links

PROCEDURE

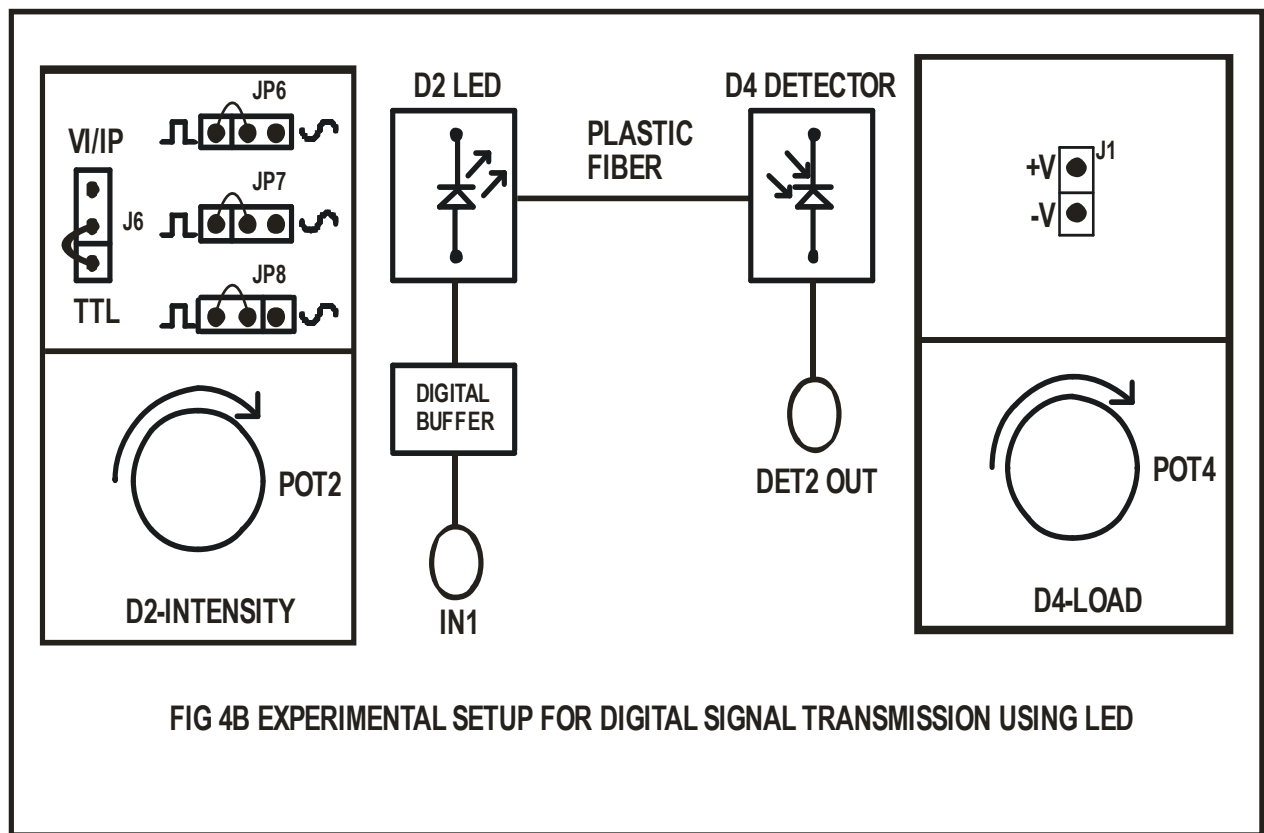
A) Digital Signal transmission using 1310nm LASER and InGaAs photo detector.

- Refer to fig 4a and carry out the connections
- Keep Intensity Control pot POT1 fully anticlockwise.
- Keep jumper J6 and JP9 to TTL mode.
- Connect the power supply of FOL-M-GP and switch it ON.
- Apply TTL signal of frequency 1 KHz from function generator to the IN1 post of Digital buffer.
- Connect Optical O/P of Laser D1 to photo diode D3 using 1 meter single mode glass fiber patch chord provided.
- Vary the Intensity control Pot POT1 and observe the detected signal at DET1 OUT post.
- Go on increasing the Function Generator TTL frequency in steps of 1 KHz and observe the Detected Signal.
- To get the undistorted detected signal adjust load using POT 3.

OBSERVATIONS:

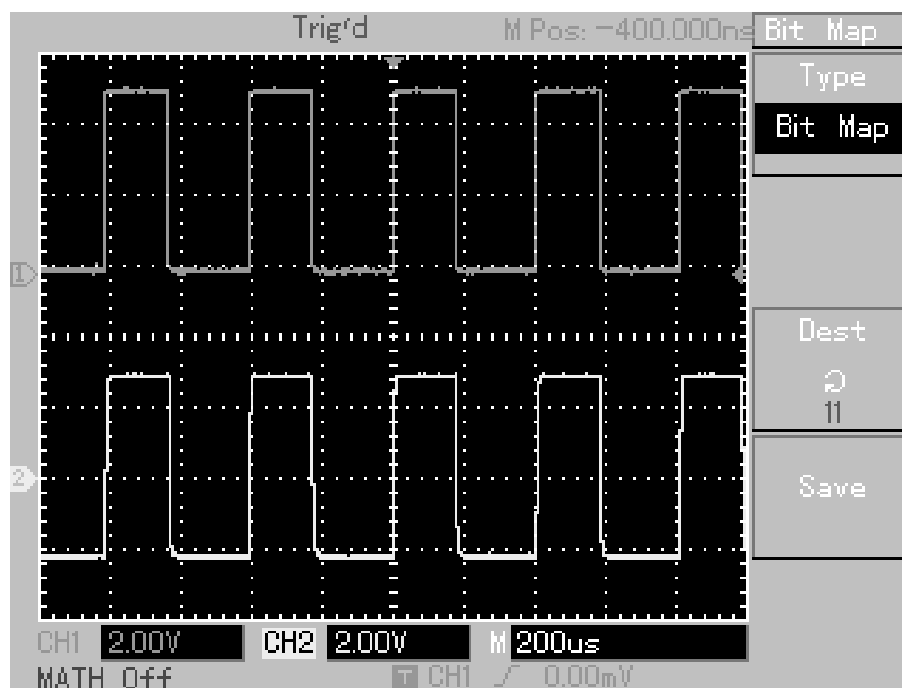
CH1: Input signal from Function generator

CH2: Detected signal through fiber optic link



B) Digital Signal transmission using 660nm LED and PIN photo transistor.

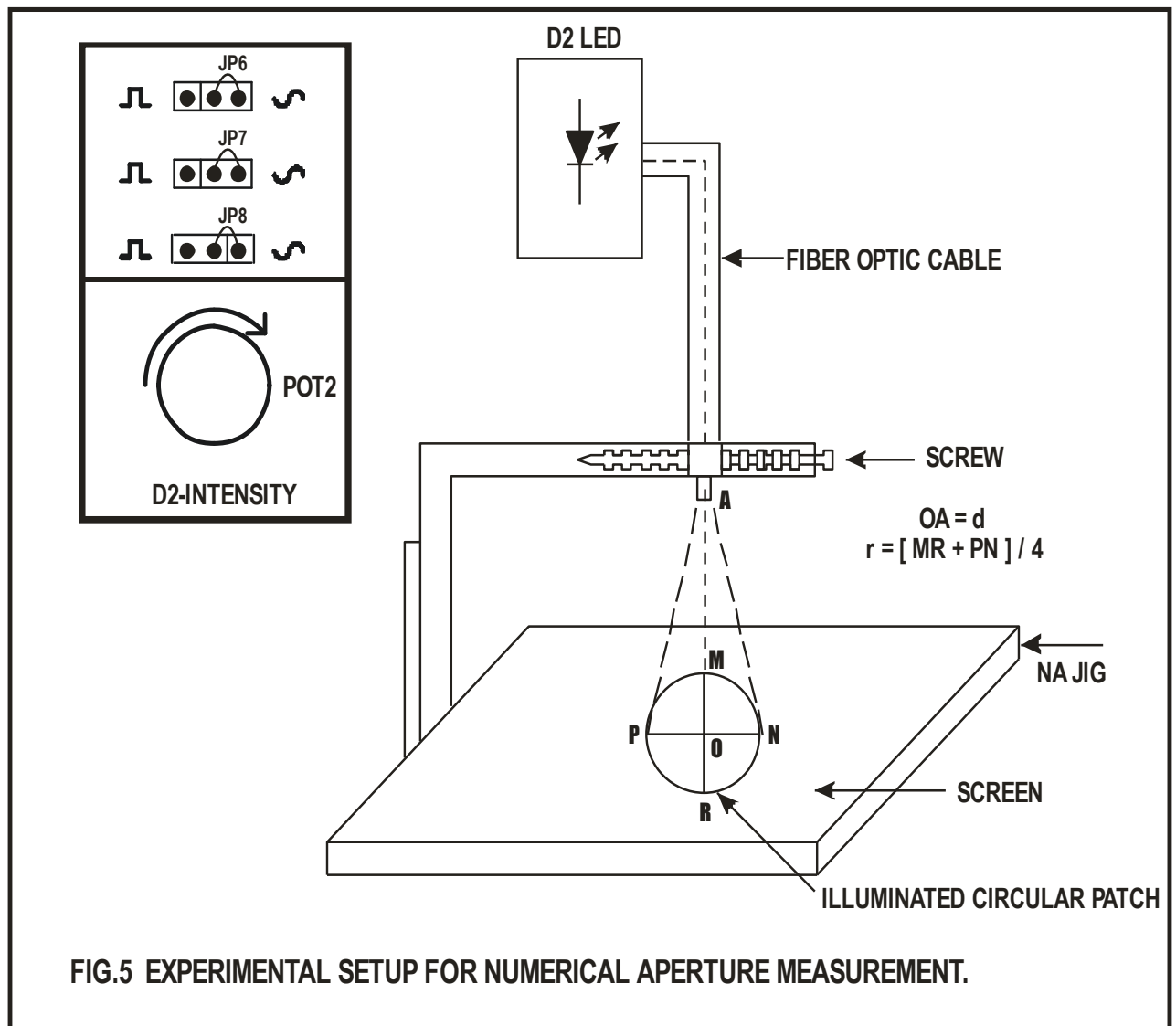
- Refer to fig 4b and carry out the connections
- Keep Intensity Control pot POT2 fully anticlockwise.
- Keep jumper J6 to TTL mode.
- Keep jumper JP6, JP7 and JP8 to Digital mode.
- Connect the power supply of FOL-M-GP and switch it ON.
- Apply TTL signal of frequency 1 KHz from function generator to the IN1 post of Digital buffer.
- Connect Optical O/P of LED D2 to photo transistor D4 using 1 meter plastic fiber patch chord provided.
- Vary the Intensity control Pot POT2 and observe the detected signal at DET2 OUT post.
- Go on increasing the Function Generator TTL frequency in steps of 1 KHz and observe the Detected Signal.
- To get the undistorted detected signal adjust load using POT 4.

OBSERVATION:

CH1: Input signal from Function generator

CH2: Detected signal through fiber optic link

EXPERIMENT NO.5



EXPERIMENT NO.5

NAME

Study of Numerical Aperture of Optical Fiber

OBJECTIVE

The objective of this experiment is to measure the numerical aperture of the plastic Fiber provided with the kit using 660nm wavelength LED.

EQUIPMENTS

- FOL-M-GP
- Power supply
- 1 meter plastic fiber
- NA Jig

THEORY

Numerical aperture refers to the maximum angle at the light incident on the fiber end is totally internally reflected and is transmitted properly along the Fiber. The cone formed by the rotations of this angle along the axis of the Fiber is the cone of acceptance of the Fiber. The light ray should strike the fiber end within its cone of acceptance; else it is refracted out of the fiber core.

Consideration in A Measurement

- It is very important that the source should be properly aligned with the cable & the distance from the launched point & the cable be properly selected to ensure that the maximum amount of Optical Power is transferred to the cable.
- This experiment is best performed in a less illuminated room.

PROCEDURE

- Make connections as shown in FIG5. Connect the power supply with proper polarity to FOL-M-GP Kit. While connecting this, ensure that the power supply is OFF.
- Keep Intensity control pot POT2 towards minimum position.
- Keep jumpers JP6, JP7 and JP8 to analog mode.
- Switch ON the power supply.

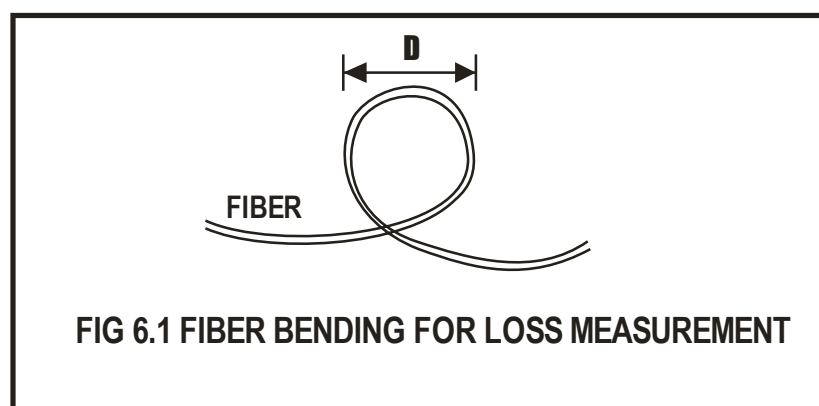
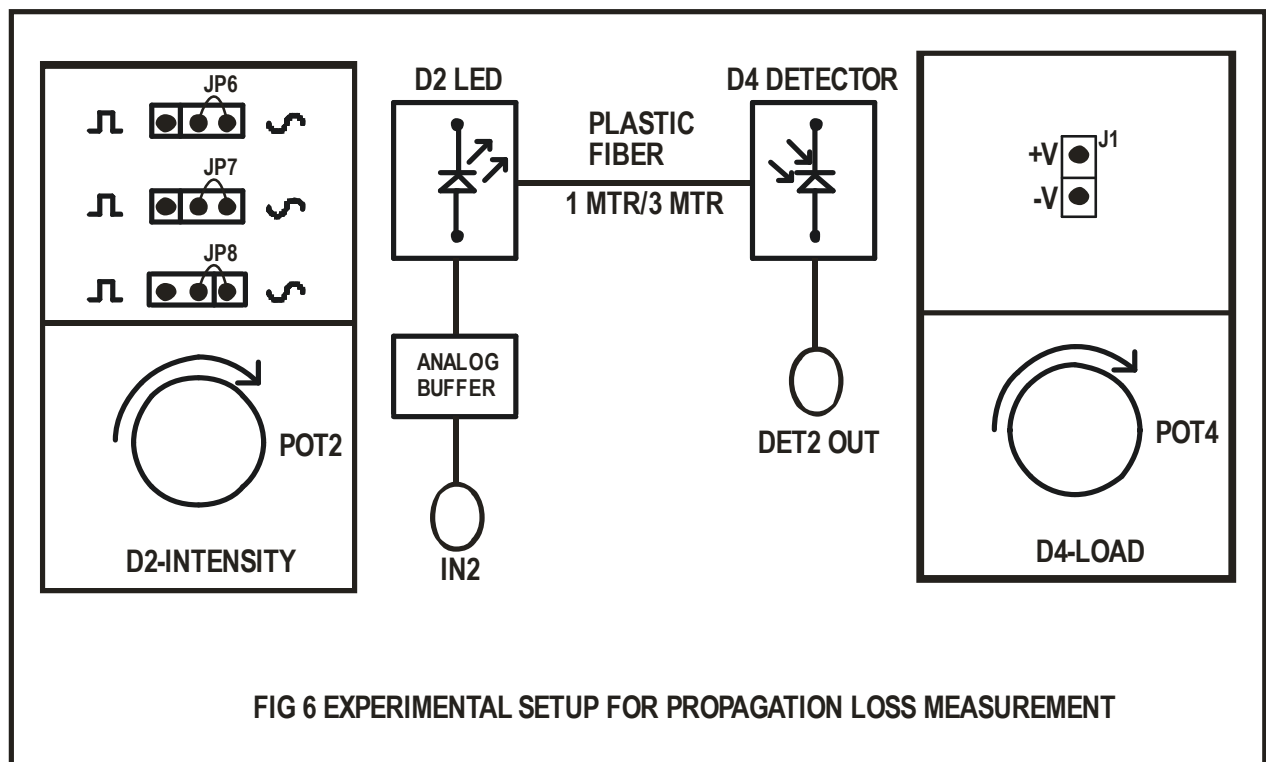
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1 Meter Fiber into the cap. Now tighten the cap by screwing it back.
- Insert the other end of the Fiber into the numerical aperture measurement jig. Adjust the fiber such that its cut face is perpendicular to the axis of the Fiber.
- Keep the distance of about 5mm between the fiber tip and the screen. Gently tighten the screw and thus fix the fiber in the place.
- Increase the intensity pot POT2 to get bright red light circular patch.
- Now observe the illuminated circular patch of light on the screen.
- Measure exactly the distance **d** and also the vertical and horizontal diameters **MR** and **PN** as indicated in the FIG 5.
- Mean radius is calculated using the following formula

$$r = (MR+PN)/4$$
- Find the numerical aperture of the Fiber using the formula

$$NA = \sin\theta_{\max} = r / \sqrt{d^2 + r^2}$$

Where $\theta_{\max}$ is the maximum angle at which the light incident is properly transmitted through the fiber.

EXPERIMENT NO.6



EXPERIMENT NO. 6

NAME

Study of Losses in Optical Fiber

OBJECTIVE

The objective of this experiment is to measure propagation loss & bending losses for two different wavelengths in plastic Fiber provided with the kit.

EQUIPMENTS

- FOL-M-GP
- Power supply
- 1 meter and 3 meter plastic fiber
- DSO
- Function generator
- Connecting Links

THEORY

Optical Fibers are available in different variety of materials. These materials are usually selected by taking into account their absorption characteristics for different wavelengths of light. In case of Optical Fiber, since the signal is transmitted in the form of light, which is completely different in nature as that of electrons, one has to consider the interaction of matter the radiation to study the losses in fiber.

Losses are introduced in fiber due to various reasons. As light propagates from one end of Fiber to another end, part of it is absorbed in the material exhibiting absorption loss. Also part of the light is reflected back or in some other directions from the impurity particles present in the material contributing to the loss of the signal at the other end of the Fiber. In general terms it is known as propagation loss. Plastic Fibers have higher loss of the order of 180 dB/Km

Whenever the condition for angle of incidence of the incident lights is violated the losses are introduced due to refraction of light. This occurs when fiber is subjected to bending. Lower the radius of curvature more is the loss. Other losses are due to the coupling of Fiber at LED & photo detector ends.

PROCEDURE

- Refer to fig 6 and carry out the connections
- Keep Intensity Control pot POT2 fully anticlockwise.
- Keep jumper JP6, JP7 and JP8 to analog mode.
- Connect the power supply with proper polarity to FOL-M-GP Kit.
- Apply 2Vp-p sine wave of frequency 1 KHz from function generator to the IN2 post of analog buffer.
- Slightly unscrew the cap of diode D2, SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1 Meter Fiber into the cap. Now tighten the cap by screwing it back.
- Connect the other end of the Fiber to detector D4, SFH350V (Photo Transistor Detector) very carefully as per the instructions in above step.
- Observe the detected signal at post DET2 OUT on oscilloscope. Adjust Intensity control pot POT2 Optical Power control potentiometer so that you receive signal of 2Vpp amplitude.
- Measure the peak value of the received signal at DET2 OUT post. Let this value be V_1 .
- Now replace 1 meter Fiber by 3 Meter Fiber between same LED and Detector. Do not disturb any settings. Again take the peak voltage reading and let it be V_2 .
- If α is the attenuation of the Fiber then we have.

$$\alpha_{dB} = (10/L1-L2) \log_{10} (V_2/V_1)$$

Where

α = dB / Km,

L_1 = Fiber Length for V_1

L_2 = Fiber Length for V_2

This α is for peak wavelength of 660nm

Measurement of Bending Losses

- Set up the 660 nm analog link using 1-meter fiber as per procedure above.
- Bend the Fiber in a loop. (As shown in FIG. 6.1) measure the amplitude of the received signal.
- Keep reducing the diameter of bend to about 2 cm & take corresponding out voltage readings. (Do not reduce loop diameter less than 1 cm).
- Plot a graph of the received signal amplitude versus the loop diameter.

**EXPERIMENT
NO.7**

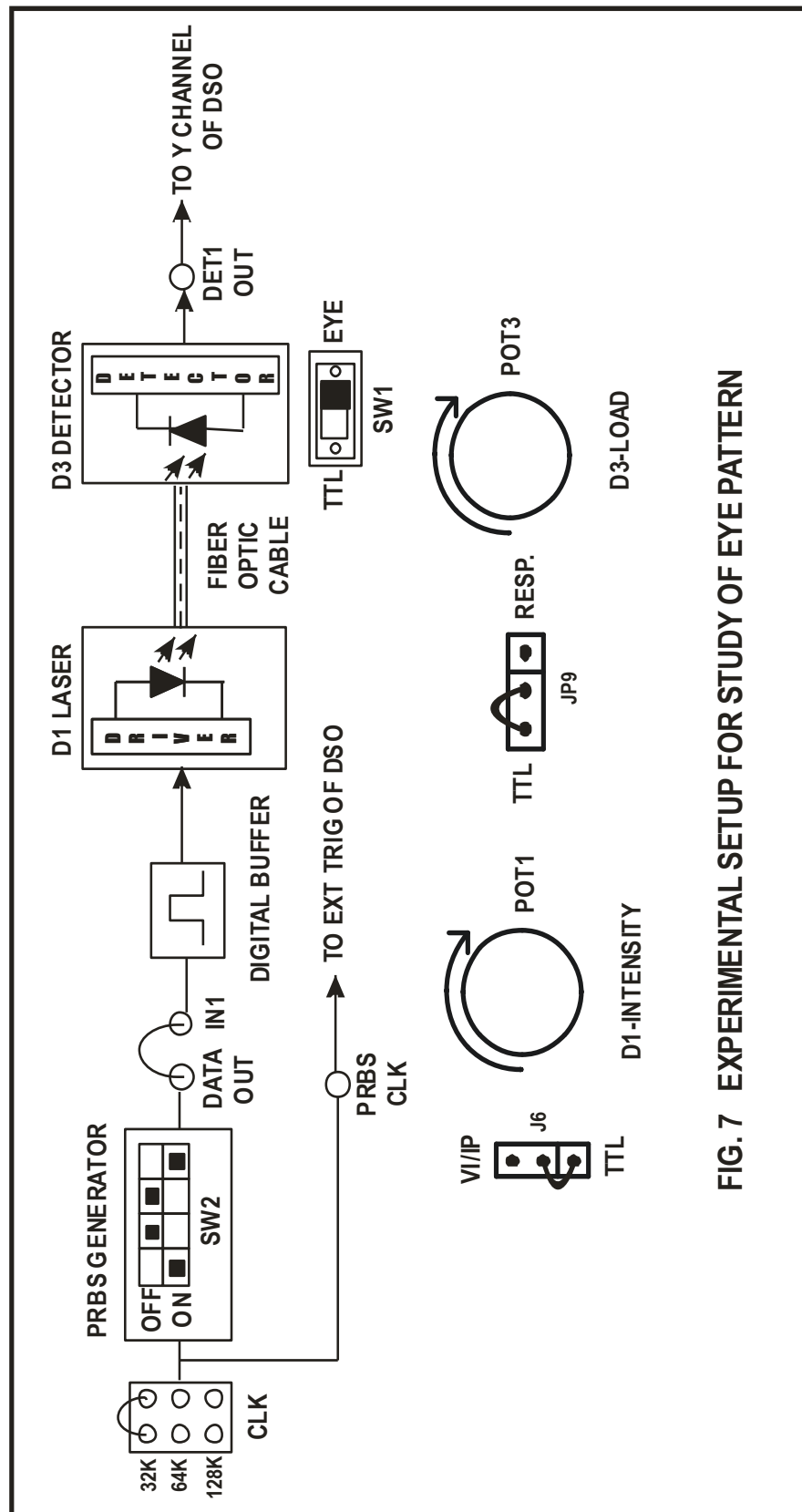


FIG. 7 EXPERIMENTAL SETUP FOR STUDY OF EYE PATTERN

EXPERIMENT NO. 7

NAME

To estimate the bandwidth of given fiber link

OBJECTIVE

To estimate bandwidth of link using

- A) EYE pattern method
- B) Rise time calculation of the link

EQUIPMENTS

- FOL-M-GP
- Power supply
- 1 meter glass fiber
- DSO
- Function Generator
- Connecting Links

THEORY

The eye-pattern technique is a simple but powerful measurement method for assessing the data-handling ability of a digital transmission system. This method has been used extensively for evaluating the performance of wire systems and can also be applied to optical fiber data links. The eye-pattern measurements are made in the time domain and allow the effects of waveform distortion to be shown immediately on an oscilloscope.

An eye-pattern can be observed with the basic equipment shown in Fig.7. The output from a pseudorandom data pattern generator is applied to the vertical input of an oscilloscope and the data rate is used to trigger the horizontal sweep. This results in the type of pattern shown in Fig. 12.2, which is called the **eye pattern** because the display shape resembles a human eye. To see how the display pattern is formed, consider the eight possible 4-bit-long NRZ combinations. When these sixteen combinations are superimposed simultaneously, an eye pattern as shown in Fig. 12.2 is formed.

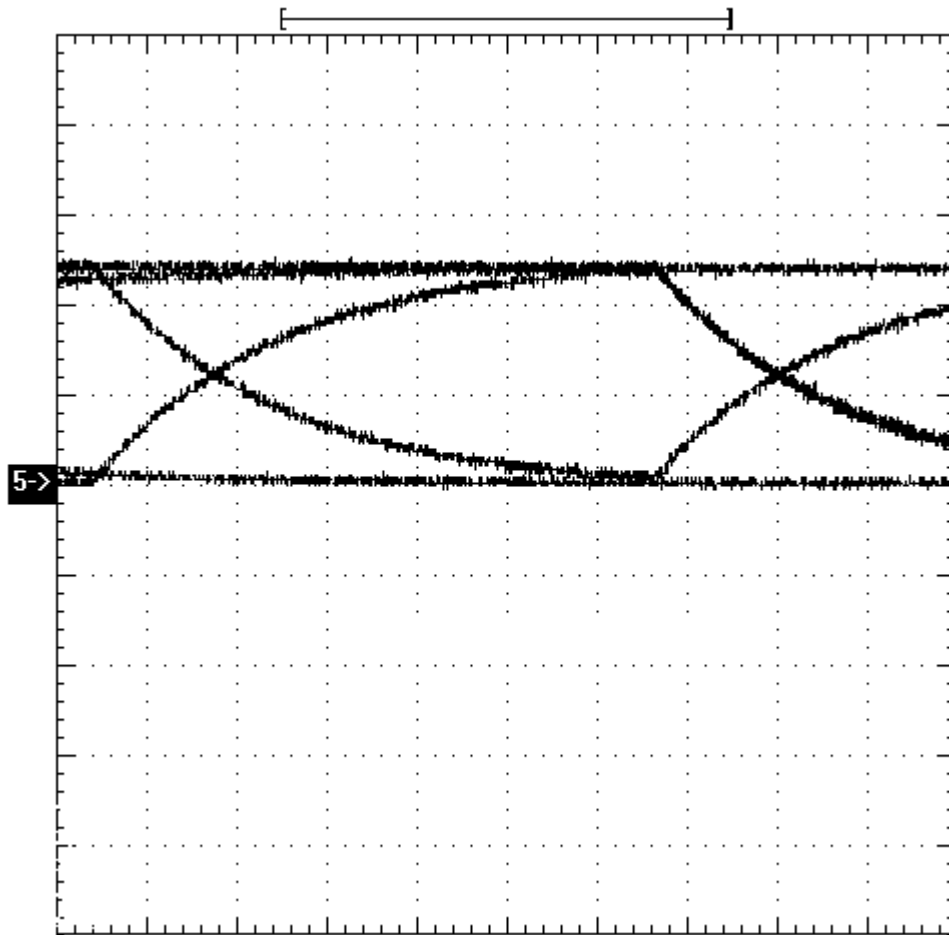
To measure system performance with the eye-pattern method, a variety of word patterns should be provided. A convenient approach is to generate a random data signal, because this is the characteristic of data streams found in practice. This type of signal generates ones and zeros at a uniform rate but in a random manner. A variety of pseudorandom pattern generators are available for this

purpose. The word pseudorandom means that the generated combination or sequence of ones and zeros will eventually repeat but that it is sufficiently random for test purposes. A pseudorandom bit sequence comprises four different 2-bit-long combinations, eight different 3-bit-long combinations, sixteen different 4-bit-long combinations and so on (that is, sequences of different N-bit-long combinations) up to a limit set by the instrument. After this limit has been generated, the data sequence will repeat.

A great deal of system performance information can be deduced from the eye-pattern display. To interpret the eye pattern, follow the procedure ahead.

PROCEDURE

- Make connections as shown in fig.7. Connect the power supply with proper polarity to FOL-M-GP Kit. While connecting this, ensure that the power supply is OFF.
- Keep switch SW2 as shown in fig 7 to generate PRBS signal.
- Keep the jumper setting as shown in fig 7.
- Keep the switch SW1 at fiber optic detector D3 to EYE position.
- Select PRBS generator clock at 32 KHz by keeping jumper CLK at first position.
- Keep Jumper J6 towards TTL position.
- Switch ON the power supply.
- Select PRBS generator clock at 32 KHz by keeping CLK jumper at first position.
- Switch ON the power supply.
- Connect the post DATA OUT of PRBS Generator to the IN1 post of digital buffer.
- Connect laser output of D1 to detector D3 using 1 meter single mode patch chord provided.
- Connect PRBS CLK of PRBS Generator to EXT. TRIG. Of oscilloscope.
- Connect detected signal **DET1 OUT** to vertical channel **Y** input of oscilloscope. Then observe EYE PATTERN by selecting EXT. TRIG KNOB on oscilloscope. Observe the Eye pattern for different clock frequencies. As clock frequency increases the EYE opening becomes smaller.

OBSERVATION:

EXPERIMENT NO.8

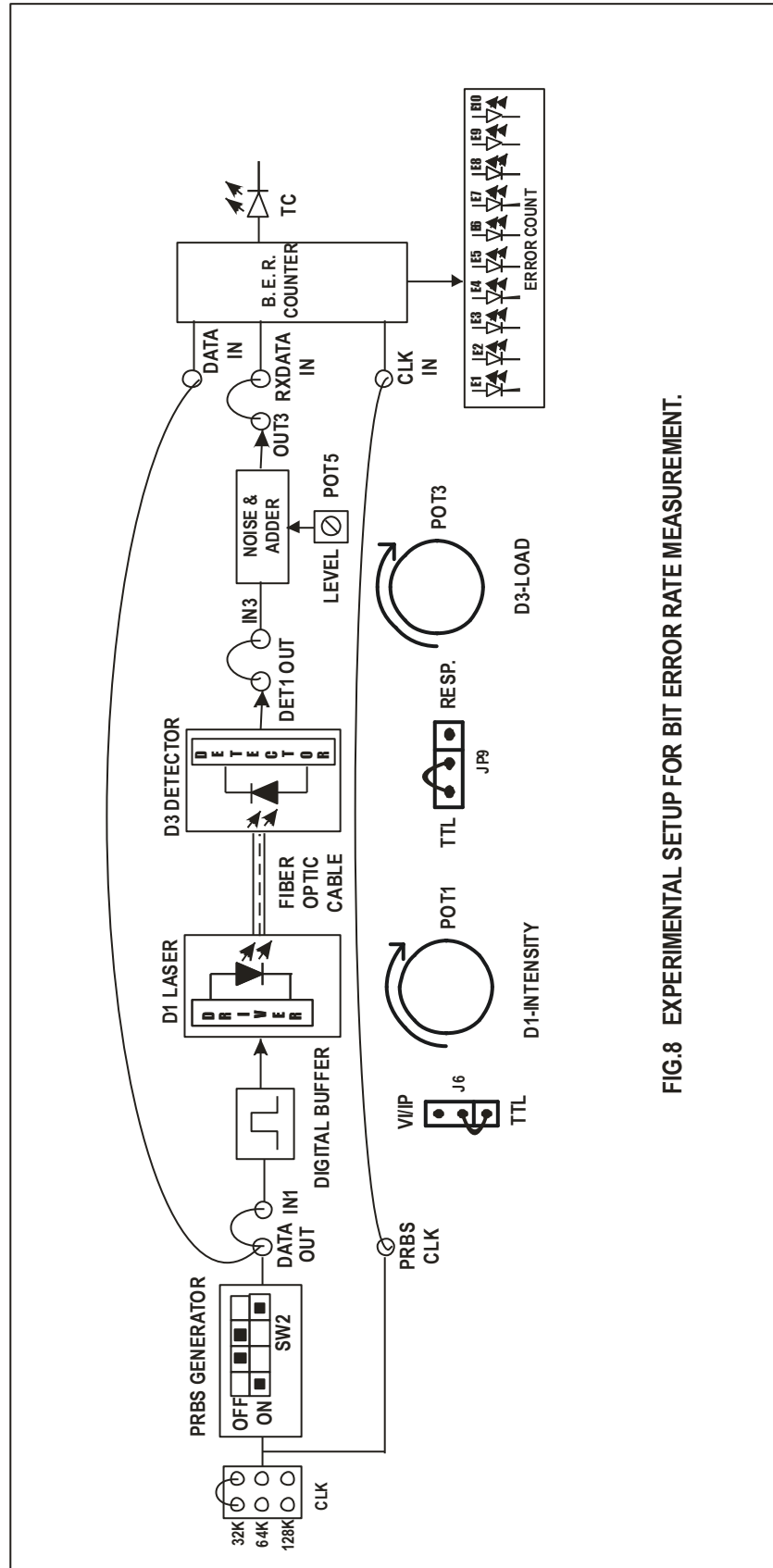


FIG.8 EXPERIMENTAL SETUP FOR BIT ERROR RATE MEASUREMENT.

EXPERIMENT NO. 8

NAME

To study and measurement of BER.

OBJECTIVE

Measurement of BER using 1310nm LASER.

EQUIPMENTS

- FOL-M-GP
- Power supply
- 1 meter glass fiber patch chord
- DSO
- Connecting Links

THEORY

Bit Error Rate

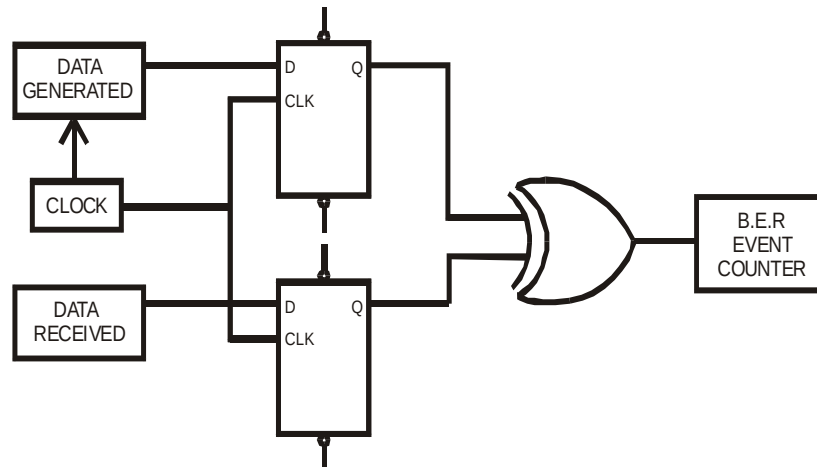
In telecommunication transmission, the bit error rate (BER) is a Ratio of bits that have errors relative to the total number of bits received in a transmission. The BER is an indication of how often a packet or other data unit has to be retransmitted because of an error. Too high BER may indicate that a slower data rate would actually improve overall transmission time for a given amount of transmitted data since the BER might be reduced, lowering the number of packets that had to be resent.

Measuring Bit Error Rate

A BERT (bit error rate tester) is a procedure or device that measures the BER For a given transmission. The BER or quality of the digital link is calculated from the number of bits received in error divided by the number of bits transmitted.

$$\text{BER} = (\text{Bits in Error}) / (\text{Total bits transmitted})$$

Using a bench test setup, this is easily measured by means of a comparator in which the transmitted bits are matched in an XOR gate with the received bits. Fig shows the schematic of the device used for the following measurements.



BIT ERROR RATE TESTER (B.E.R.T.)

If the bits are alike at the XOR gate input, when clocked in from the D flip flop, the output is low. If they are different, the XOR output goes high, causing an event count. The event counter can be set for various time periods. In general, the longer the time period, the more accurate is the count.

A random character generator and white noise source should be used for these measurements.

The number of bit errors is dependent upon the amount of noise entering the system. White noise or background noise has an average or RMS value that is exceeded periodically by peaks that rise many times that level. These peaks exist only for a very short period of time. When the peak equals or exceeds the signal level, that is noise energy = bit energy, there is a 50/50 chance of error. The peak time periods can be calculated statistically from the error function.

In Link-B, PRBS sequence is generated by using a 4-bit right shift register whose feedback is completed by the EX-OR gate.

Let Initially 1001 be the 4-bit switch setting on the SW7.

Clock States	D1	D2	D3	D4	C
			A	B	C
1	1	0	0	1	1
2	1	1	0	0	0
3	0	1	1	0	1
4	1	0	1	1	0
5	0	1	0	1	1
6	1	0	1	0	1
7	1	1	0	1	1
8	1	1	1	0	1
9	1	1	1	1	0
10	0	1	1	1	0
11	0	0	1	1	0
12	0	0	0	1	1
13	1	0	0	0	0

14	0	1	0	0	0
15	0	0	1	0	1
16	1	0	0	1	1

Thus the sequence repeats constantly with a period corresponding to 16 clock states.

Length of sequence = $2^4 = 16$

Now the Pseudo Random Sequence pattern is C = 1010111100010011

PROCEDURE

- Make connections as shown in fig.8. Connect the power supply with proper polarity to FOL-M-GP Kit. While connecting this, ensure that the power supply is OFF.
- Keep switch SW2 as shown in fig 8 to generate PRBS signal.
- Keep the jumper setting as shown in fig 8.
- Keep the switch SW1 at fiber optic detector D3 to TTL position.
- Select PRBS generator clock at 32 KHz by keeping jumper CLK at first position.
- Keep Jumper J6 towards TTL position.
- Switch ON the power supply.
- Connect the post DATA OUT of PRBS Generator to the IN1 post of Digital Buffer and also to the DATA IN post of BER COUNTER section
- Connect laser output of D1 to detector D3 using 1 meter single mode patch chord provided.
- Connect detected signal DET1 OUT to post IN3 of Noise Source.
- Connect post OUT3 of Noise Source to post RXDATA IN of Bit Error Rate event counter.
- Connect post PRBS CLK of PRBS Generator to post CLK IN of BER counter.
- Press Switch above IN1 to start counter.
- Vary pot POT5 for **Noise Level** to observe effect of noise level on the error count.
- Observe the Error Count LEDs for the error count in received signal in time 10 seconds.

BER Measurement

As per the definition the BER is a ratio of Errored bits (E_b) to total bits Transmitted (T_b) in a period of time t seconds.

i.e. $BER = E_b / T_b$

For eg. In this experiment if PRBS data is transmitted at 32Kbits per second (i.e. jumper selection at 32 KHz) for a period of 10 seconds.

So total bits transmitted in 10 seconds (T_b) = 320Kbits.

The TTL OUT data & data with noise is fed to BER counter which compares the two data inputs at each clock input.

The counter displays the Error count (Eb) on LED in 10-bit binary form (e.g. 0000001010), which has to be converted in decimal form (it becomes 10) so the BER ratio then becomes

$$\begin{aligned}\text{BER} &= 10 / (320 \times 10^3) \\ &= 0.00003125\end{aligned}$$

i.e. the channel Bit Error Rate ratio is 3.1×10^{-5} ($3 / 100000$) or in other words we can say that out of 100000 bits transmitted through the channel the channel gives 3 bits in error.